THE COLLATZ CONJECTURE & NON-ARCHIMEDEAN SPECTRAL THEORY - PART II - (p,q)-ADIC FOURIER ANALYSIS AND WIENER'S TAUBERIAN THEOREM

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Abstract. This paper gives an overview of (p,q)-adic Fourier theory—the Fourier theory of functions from the p-adic numbers to the q-adic numbers, where p and q are distinct primes—which we then use to prove a novel (p,q)-adic generalization of Norbert Wiener's celebrated Tauberian Theorem. Letting K be a metrically complete, algebraically closed local field of residue characteristic q, letting $C(\mathbb{Z}_p,K)$ be the Banach space of continuous functions $\mathbb{Z}_p \to K$, and letting $d\mu$ be a (p,q)-adic measure (a continuous linear functional $C(\mathbb{Z}_p,K) \to K$), the (p,q)-adic Wiener Tauberian Theorem (WTT) we prove establishes the equivalence of the density of the span of translates of $d\mu$'s Fourier-Stieltjes Transform and the non-vanishing of the Radon-Nikodym derivative of $d\mu$ at all points in $\mathbb{Z}_p$ where the derivative exists in K.

Contents

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1. Preliminaries & Introduction

Preliminaries. A good deal of notation is involved in this paper. We cover it in full at the beginning of **Section 2**. For now, we just state the bare minimum required for the paper's introduction.

Let p and q be primes. Unless explicitly stated otherwise, we always assume $p \neq q$. In the spirit of number theory, we also allow q to be ∞.

Definition 1. By a ***q*-adic field**, we mean a local field equipped with an absolute value, denoted $\left|\cdot\right|_q$, with respect to which the field is complete as a metric space. If $q=\infty$, we require $\left|\cdot\right|_q$ to be an archimedean absolute value. If $q<\infty$, we require $\left|\cdot\right|_q$ to be a non-archimedean absolute value, and require the q-adic field to have a residue field of characteristic q, so that $\left|p\right|_q=1$ for all prime numbers $p\neq q$. If $q=\infty$, we write $\left|\cdot\right|_q$ as $\left|\cdot\right|_\infty$.

Notation 1. For any prime p, we write $\mathbb{Z}_p$ and $\mathbb{Q}_p$ to denote the ring of p-adic integers and fields of p-adic rational numbers, respectively, equipped with the standard p-adic absolute value $\left|\cdot\right|_p$, with $\left|p\right|_p=1/p$. We write $\mathbb{C}_p$ to denote the field of p-adic complex numbers, i.e., the metric completion of the algebraic closure of $\mathbb{Q}_p$ We write $v_p\left(\cdot\right)$ to denote the p-adic valuation, with $\left|\cdot\right|_p=p^{-v_p(\cdot)}$ and $v_p\left(0\right)\overset{\text{def}}{=}\infty$, where $\overset{\text{def}}{=}$ means "by definition".

We write $\hat{\mathbb{Z}}_p$ to denote $\mathbb{Z}\left[1/p\right]/\mathbb{Z}$, the Pontryagin dual of $\mathbb{Z}_p$. We identify $\hat{\mathbb{Z}}_p$ with the group of rational numbers in $[0,1)$ whose denominators are powers of p. We also embed $\hat{\mathbb{Z}}_p$ in $\mathbb{Q}_p$, where it is isomorphic to the quotient $\mathbb{Q}_p/\mathbb{Z}_p$. For any $t\in\hat{\mathbb{Z}}_p\backslash\left\{0\right\}$, observe that t can be written in irreducible form as k/p^n for some integers $n\geq 1$ and some $k\in\{0,\ldots,p^n-1\}$ which is co-prime to p. Consequently, the p-adic absolute value of t is $\left|t\right|_p=p^n$, and the numerator of t is given by $t\left|t\right|_p=k$, with both values being 0 when $t=0$.

For aesthetic reasons, we write p-adic variables in lower-case $\mathfrak{fraktur}$ font.

When a limit is being taken in a metric space, we write the metric space in which the convergence occurs over the equals sign. Hence:

$$\lim_{n\rightarrow\infty}f\left(n\right)\overset{\mathbb{R}}{=}0 \tag{1.1}$$

means that f converges to 0 in the reals, whereas:

$$\lim_{n\rightarrow\infty}f\left(n\right)\overset{\mathbb{Q}_5}{=}0 \tag{1.2}$$

means that f converges to 0 in the 5-adics.

Finally, letting (M,d) be a metric space, letting L be a point of M, and considering a function $\varphi:\hat{\mathbb{Z}}_p\rightarrow M$ (where $\hat{\mathbb{Z}}_p$ is the pontryagin dual of $\mathbb{Z}_p$) we say that $\varphi\left(t\right)$ **converges (in M) to L as $\left|t\right|_p\rightarrow\infty$**, written:

$$\lim_{\left|t\right|_p\rightarrow\infty}\varphi\left(t\right)\overset{M}{=}L \tag{1.3}$$

when, for all $\epsilon>0$, there exists an $N\geq 1$ which makes $d\left(\varphi\left(t\right),L\right)<\epsilon$ for all $t\in\hat{\mathbb{Z}}_p$ with $\left|t\right|_p\geq p^N$. Thus, for example, given $\hat{\chi}:\hat{\mathbb{Z}}_p\rightarrow\mathbb{C}_q$, we say $\lim_{\left|t\right|_p\rightarrow\infty}\left|\hat{\chi}\left(t\right)\right|_q\overset{\mathbb{R}}{=}0$ to mean that, for all $\epsilon>0$, there exists an $N\geq 1$ so that the real number $\left|\hat{\chi}\left(t\right)\right|_q$ is less than ϵ for all $t\in\hat{\mathbb{Z}}_p$ with $\left|t\right|_p\geq p^N$.

Lastly, before we can proceed, we need to review some basic terminology of non-archimedean functional analysis.

Definition 2. Let $\mathbb{F}$ be a q-adic field. We write $C(\mathbb{Z}_p,\mathbb{F})$ to denote the vector space of **continuous** (p,q)**-adic** functions $\chi:\mathbb{Z}_p\to\mathbb{F}$. We make $C(\mathbb{Z}_p,\mathbb{F})$ into a non-archimedean Banach space by equipping it with the supremum norm:

$$\|\chi\|_{\mathbb{Z}_p,q}\overset{\text{def}}{=}\sup_{\mathfrak{z}\in\mathbb{Z}_p}|\chi(\mathfrak{z})|_q \tag{1.4}$$

We write $B(\mathbb{Z}_p,\mathbb{F})$ to denote the Banach space of bounded (p,q)-adic functions. This space has $\|\cdot\|_{\mathbb{Z}_p,q}$ as its norm and contains $C(\mathbb{Z}_p,\mathbb{F})$ as a closed subspace.

Similarly, we write $B\left(\hat{\mathbb{Z}}_p,\mathbb{F}\right)$ to denote the non-archimedean Banach space of all q-adically bounded functions $\hat{\mathbb{Z}}_p\to\mathbb{F}$:

$$B\left(\hat{\mathbb{Z}}_p,\mathbb{F}\right)\overset{\text{def}}{=}\left\{\hat{\chi}:\hat{\mathbb{Z}}_p\to\mathbb{F}\text{ so that }\sup_{t\in\hat{\mathbb{Z}}_p}|\hat{\chi}(t)|_q<\infty\right\} \tag{1.5}$$

Its norm is:

$$\|\hat{\chi}\|_{\hat{\mathbb{Z}}_p,q}\overset{\text{def}}{=}\sup_{t\in\hat{\mathbb{Z}}_p}|\hat{\chi}(t)|_q \tag{1.6}$$

We write $c_0\left(\hat{\mathbb{Z}}_p,\mathbb{F}\right)$ to denote the non-archimedean Banach space of all functions $\hat{\mathbb{Z}}_p\to\mathbb{F}$ which decay to 0 at ∞:

$$c_0\left(\hat{\mathbb{Z}}_p,\mathbb{F}\right)\overset{\text{def}}{=}\left\{\hat{\chi}\in B\left(\hat{\mathbb{Z}}_p,\mathbb{F}\right):\lim_{|t|_p\to\infty}|\hat{\chi}(t)|_q\overset{\mathbb{R}}{=}0\right\} \tag{1.7}$$

Note that $c_0\left(\hat{\mathbb{Z}}_p,\mathbb{F}\right)$ is a closed subspace of $B\left(\hat{\mathbb{Z}}_p,\mathbb{F}\right)$.

Remark 1. When there is no risk of confusion due to our systematic use of $\widehat{\text{hats}}$, we will write $\|\cdot\|_{p,q}$ to denote the supremum norms $\|\cdot\|_{\mathbb{Z}_p,q}$ and $\|\cdot\|_{\hat{\mathbb{Z}}_p,q}$. As a rule, if the function in the norm has a hat on it, the supremum is over $\hat{\mathbb{Z}}_p$; else, it is over $\mathbb{Z}_p$. Thus:

$$\|\chi\|_{p,q}\overset{\text{def}}{=}\sup_{\mathfrak{z}\in\mathbb{Z}_p}|\chi(\mathfrak{z})|_q \tag{1.8}$$

$$\|\hat{\chi}\|_{p,q}\overset{\text{def}}{=}\sup_{t\in\hat{\mathbb{Z}}_p}|\hat{\chi}(t)|_q \tag{1.9}$$

As is standard, we define measures as continuous linear functionals on the Banach space of continuous functions.

Definition 3. A (p,q)**-adic measure** is a continuous linear functional $C(\mathbb{Z}_p,\mathbb{F})\to\mathbb{F}$. The space of all such measures is denoted $C(\mathbb{Z}_p,\mathbb{F})'$; this is the continuous dual of $C(\mathbb{Z}_p,\mathbb{F})$. Given a measure $d\mu\in C(\mathbb{Z}_p,\mathbb{F})'$, we write:

$$\int_{\mathbb{Z}_p}f(\mathfrak{z})\,d\mu(\mathfrak{z}) \tag{1.10}$$

to denote the image of a function $f\in C(\mathbb{Z}_p,\mathbb{F})$ under $d\mu$. $C(\mathbb{Z}_p,\mathbb{F})'$ is a non-archimedean Banach space under the **total variation norm**:

$$\|d\mu\|\overset{\text{def}}{=}\sup_{\substack{f\in C(\mathbb{Z}_p,\mathbb{F})\\ \|f\|_{p,q}\leq 1}}\left|\int_{\mathbb{Z}_p}f(\mathfrak{z})\,d\mu(\mathfrak{z})\right|_q \tag{1.11}$$

Introduction. As we saw in Part I of this series ([24]), we can reformulate the study of the periodic points and divergent points of a Hydra map H by considering the integer values attained by H's numen χ_H. Because the author's theory of frames (see [22]) is required in order to make sense of χ_H as a function for an arbitrary Hydra map H, the reader is encouraged to think of χ_H as a function of a p-adic integer variable that produces numbers as output, though in simple cases, such as when H is the Shortened $qx+1$ map, T_q, χ_H will take values in the q-adics for some prime q, and therefore be an element of $B(\mathbb{Z}_p, \mathbb{Q}_q)$. Because of this, it is natural to use the methods of (p,q)-adic analysis to study χ_H and the values it takes. Better still, these methods also apply to the general case.

Classical value distribution theory is intimately connected to the theory of analytic functions. In complex analysis, **Nevanlinna theory** uses **Jensen's formula** to study the value distribution of meromorphic functions on $\mathbb{C}$ [3, 4]. Aside from its use in analysis, this circle of ideas is also important in number theory, where there is a deep connection between Nevanlinna theory and the theory of heights in Diophantine approximation, thanks to the well-known algebraic properties common to both number fields and fields of meromorphic functions [4]. Because value distribution theory is so closely connected to analytic function theory, it is unsurprising that Nevanlinna theory and Jensen's formula have been generalized to the study of analytic functions over $\mathbb{Q}_p$ or any metrically complete extension thereof [3]. Unfortunately, all of this theory falls apart in the case of (p,q)-adic functions for distinct primes p and q.

Analytic functions are those that are locally defined by convergent power series, a construction not available in (p,q); it makes no sense to consider a polynomial in a p-adic variable with q-adic coefficients. However, while (p,q)-adic function theory might not have power series and analytic functions, it—much like the study of real- or complex-valued functions of a p-adic variable—(p,q)-adic analysis *does* have a workable theory of Fourier analysis. Not only can this Fourier theory be used to study (p,q)-adic functions in greater detail, it also provides an alternative route to a theory of value distribution by way of spectral theory.

As was explained in the introduction of [24], in a Banach algebra A of functions $f : X \to K$ where X is some set and K is a valued field, the **spectrum** of a function $f \in A$ is precisely the set of values f attains in K. Thus, the value distribution theory of f is precisely the study of those scalars $\lambda \in K$ for which the function $f(x) - \lambda$ is not a unit of A; it is a unit if and only if λ is not in the image of f. Better still, in situations like ours where it makes sense of speak of the Fourier transform of a function in A, we can use Fourier analysis to attack the question of whether or not $f(x) - \lambda$ is a unit.

One of the fundamental properties of the Fourier transform is that it turns point-wise multiplication of functions into convolution:

$$\mathscr{F}\{f \times g\}(\xi) = \left(\hat{f} * \hat{g}\right)(\xi) \tag{1.12}$$

This result, the famous **Convolution theorem**, implies that $\mathscr{F}$ is an isomorphism of Banach algebras. Consequently, $\mathscr{F}$ will send units of A to units of $\mathscr{F}(A)$, and we can use this to check whether or not there is an $x_0 \in X$ at which $f(x) = \lambda$, and the fact that this works is precisely the content of **Wiener's Tauberian Theorem**.

WTT is a classic of twentieth-century harmonic analysis, simultaneously unifying and generalizing such divers topics as divergent series, summability methods, asymptotic analysis, abstract harmonic analysis, and the theory of Banach algebras. The interested reader can consult [10] for an excellent exposition of most of these; [28] gives a more advanced treatment in the context of the theory of distributions and their applications to physics. To give the version of the WTT most relevant to this paper, we write $\mathbb{R}/\mathbb{Z}$ to denote $[0,1)$ with addition modulo 1, and write $\ell^1(\mathbb{Z},\mathbb{C})$ to denote the Banach space:

$$\ell^1(\mathbb{Z},\mathbb{C}) \overset{\text{def}}{=} \left\{\hat{\phi}:\mathbb{Z}\to\mathbb{C}:\sum_{n\in\mathbb{Z}}\left|\hat{\phi}(n)\right|<\infty\right\} \tag{1.13}$$

We write $A(\mathbb{R}/\mathbb{Z})$ to denote the **Wiener algebra on** $\mathbb{R}/\mathbb{Z}$, the Banach space of all absolutely convergent Fourier series:

$$A(\mathbb{R}/\mathbb{Z}) \overset{\text{def}}{=} \left\{t\mapsto\sum_{n\in\mathbb{Z}}\hat{\phi}(n)\,e^{2\pi int}:\hat{\phi}\in\ell^1(\mathbb{Z},\mathbb{C})\right\} \tag{1.14}$$

under the norm:

$$\|\phi\|_A \overset{\text{def}}{=} \sum_{n\in\mathbb{Z}}\left|\hat{\phi}(n)\right|,\ \forall\phi\in A(\mathbb{R}/\mathbb{Z}) \tag{1.15}$$

With this terminology, the WTT for $\ell^1(\mathbb{Z},\mathbb{C})$ can be stated as:

Theorem 1 (**Wiener's Tauberian Theorem for ℓ^1**). *Let $\phi\in A(\mathbb{R}/\mathbb{Z})$. Then, the translates of $\hat{\phi}$ are dense in $\ell^1(\mathbb{Z},\mathbb{C})$ if and only if $\phi(t)\neq 0$ for any $t\in\mathbb{R}/\mathbb{Z}$.*

In general, a WTT will exist whenever we have a pair of isomorphic Banach algebras $(A,\times)$ and $\left(\hat{A},*\right)$, one of which—as indicated—is an algebra under point-wise multiplication, while the other is an algebra under convolution. Here are some well-known examples of WTTs:

- Let $f\in L^1(\mathbb{R})$. Then, the spans of the translates of f are dense in $L^1(\mathbb{R})$ if and only if f's Fourier transform $\hat{f}$ has no real zeroes.
- Let $f\in L^2(\mathbb{R})$. Then, the spans of the translates of f are dense in $L^2(\mathbb{R})$ if and only if the set of $\hat{f}$'s real zeroes has a Lebesgue measure of 0.
- Let $\phi\in A(\mathbb{R}/\mathbb{Z})$. Then, $1/\phi\in A(\mathbb{R}/\mathbb{Z})$ if and only if ϕ has no zeroes.

As explained in the abstract, in this paper, we establish a (p,q)-adic version of the WTT. to the author's knowledge, this is a novel result. We prove two forms of this WTT, one for functions and another for measures.

Theorem 2 (**Wiener Tauberian Theorem for (p,q)-adic Functions**). *Let $q<\infty$, let K be an algebraically closed, spherically incomplete q-adic field, and let $\chi\in C(\mathbb{Z}_p,K)$. Then, the following are equivalent:*

I. $\frac{1}{\chi}\in C(\mathbb{Z}_p,K)$*;*

II. $\hat{\chi}$*, the Fourier transform, has a convolution inverse in* $c_0\left(\hat{\mathbb{Z}}_p,K\right)$*.*

III. The span of the translates of $\hat{\chi}$ *is dense in* $c_0\left(\hat{\mathbb{Z}}_p,K\right)$*;*

IV. χ *has no zeroes.*

Our second version, for measures, is:

Theorem 3 (**Wiener Tauberian Theorem for (p,q)-adic Measures**). *Let $q<\infty$, let K be an algebraically closed, spherically incomplete q-adic field, and let $d\mu\in C(\mathbb{Z}_p,K)'$. Then,* *the span of the translates of the Fourier-Stieltjes transform of $d\mu$ is dense in $c_0\left(\hat{\mathbb{Z}}_p, K\right)$ if and only if $d\mu$'s Radon-Nikodym derivative with respect to the (p,q)-adic Haar probability measure is non-zero at all points where said derivative exists in K.*

Just like in classical analysis, the spaces of measures $C(\mathbb{Z}_p, K)'$ contains an isomorphic copy of $C(\mathbb{Z}_p, K)$, and the embedding of $C(\mathbb{Z}_p, K)$ in $C(\mathbb{Z}_p, K)'$ is the map which sends a continuous (p,q)-adic function $f(\mathfrak{z})$ to the (p,q)-adic measure $f(\mathfrak{z})\,d\mathfrak{z}$, where $d\mathfrak{z}$ is the (p,q)-adic Haar probability measure. As such, **Theorem 3** implies **Theorem 2**.

The notions of the (p,q)-adic Haar probability measure and Radon-Nikodym derivative used here are explained in detail in the next section. Of the two, our Haar measure is in line with the standard (if niche) meaning of the term, however our use of Radon-Nikodym differentiation is both more specific and more general to the formulation given by Schikhof in [17] and subsequently used by other researchers, such as in [1].

2. A Brief Account of (p,q)-Adic Fourier Theory

Notation & Conventions.

Assumption 1. *UNLESS STATED OTHERWISE, FOR THE ENTIRE PAPER, WE FIX ONCE AND FOR ALL A PRIME NUMBER p AND A NUMBER q WHICH IS EITHER A PRIME OR ∞. WE ALSO FIX AN ALGEBRAICALLY CLOSED q-ADIC FIELD K. IF $q<\infty$, WE ALSO REQUIRE K TO HAVE A RESIDUE FIELD OF CHARACTERISTIC $\neq p$.*

For any real number x, we write $\mathbb{N}_x$ to denote the set of all integers $\geq x$ (thus, $\mathbb{N}_0 = \{0,1,2,\ldots\}$; $\mathbb{N}_1 = \{1,2,\ldots\}$, etc.). We write $\mathbb{Z}_p'$ to denote $\mathbb{Z}_p \backslash \mathbb{N}_0$; note that this is the set of all p-adic integers with infinitely many non-zero p-adic digits.

For any $\mathfrak{z} \in \mathbb{Z}_p$ and any integer $n \geq 0$, we write $[\mathfrak{z}]_{p^n}$ to denote the unique integer in $\{0,\ldots,p^n-1\}$ which is congruent to $\mathfrak{z}$ modulo p^n. Note then that $[\mathfrak{z}]_1 = 0$ for all $\mathfrak{z} \in \mathbb{Z}_p$. We follow Yvette Amice ([2]) in referring to the series representation $\sum_{n=-n_0}^{\infty} c_n p^n$ of a p-adic number $\mathfrak{y} \in \mathbb{Q}_p$ (be it rational or integral) as the **Hensel series** of $\mathfrak{y}$.

We employ a non-standard notation for congruences:

$$x \overset{a}{\equiv} y \\ \Updownarrow \\ x = y \mod a \tag{2.1}$$

To give some examples, for $\mathfrak{z} \in \mathbb{Z}_p$, $\mathfrak{z} \overset{p^n}{\equiv} k$ means "$\mathfrak{z}$ is congruent to k mod p^n"; i.e., $\mathfrak{z} \in k + p^n\mathbb{Z}_p$. Given $\mathfrak{x}, \mathfrak{y} \in \mathbb{Q}_p$, we write $\mathfrak{x} \overset{1}{\equiv} \mathfrak{y}$ to mean $\mathfrak{x} - \mathfrak{y} = 0 \mod \mathbb{Z}_p$. In particular, note that:

- $\mathfrak{y} \overset{1}{\equiv} \{\mathfrak{y}\}_p$ for all $\mathfrak{y} \in \mathbb{Q}_p$;
- $s, t \in \hat{\mathbb{Z}}_p$ denote the same element of $\hat{\mathbb{Z}}_p$ if and only if $s \overset{1}{\equiv} t$;
- $\mathfrak{z} \overset{1}{\equiv} k$ is true for all $\mathfrak{z} \in \mathbb{Z}_p$ and all $k \in \mathbb{Z}$.

For a rational integer n, $n \overset{2}{\equiv} 0$ means n is even, while $n \overset{2}{\equiv} 1$ means n is odd. This notation is particularly useful in conjunction with the **Iverson Bracket notation**, wherein, given a statement S, we write $[S]$ to denote the function which is 1 if S is true and 0 if S is false. Thus, for example:

$$\mathfrak{z} \mapsto \left[\mathfrak{z} \overset{p^n}{\equiv} k\right] \tag{2.2}$$

denotes the indicator function of the set $k + p^n\mathbb{Z}_p$.

We write $\{\cdot\}_p : \mathbb{Q}_p \to \hat{\mathbb{Z}}_p$ to denote the **p-adic fractional part**, with:

$$\{\mathfrak{y}\}_p \overset{\text{def}}{=} \begin{cases} 0 & \text{if } n_0 \geq 0 \\ \sum_{n=n_0}^{-1} c_n p^n & \text{if } n_0 \leq -1 \end{cases} \tag{2.3}$$

where:

$$\mathfrak{y} = \sum_{n=n_0}^{\infty} c_n p^n \tag{2.4}$$

We write $\lambda_p(n)$ to denote the number of p-adic digits of the non-negative integer n; note that $\lambda_p(n) = \left\lceil \log_p(n+1) \right\rceil$.

Given an abelian topological group G and a function $\hat{\chi} : \hat{\mathbb{Z}}_p \to G$, we write $\sum_{t \in \hat{\mathbb{Z}}_p} \hat{\chi}(t)$ to denote the limit:

$$\sum_{t \in \hat{\mathbb{Z}}_p} \hat{\chi}(t) \overset{\text{def}}{=} \lim_{N \to \infty} \sum_{|t|_p \leq p^N} \hat{\chi}(t) \overset{\text{def}}{=} \lim_{N \to \infty} \sum_{k=0}^{p^N - 1} \hat{\chi}\left(\frac{k}{p^N}\right) \tag{2.5}$$

where the limits are taken in the topology of G, provided that the limit exists.

Given $s \in \hat{\mathbb{Z}}_p$, we write $\mathbf{1}_s : \hat{\mathbb{Z}}_p \to \{0,1\}$ to denote the indicator function of $\{s\}$:

$$\mathbf{1}_s(t) \overset{\text{def}}{=} \left[t \overset{1}{\equiv} s\right] \tag{2.6}$$

Finally, we will speak of the **quality** of a field or Banach space to refer to whether the field/Banach space is archimedean or not. Thus, just as integers have a parity which is either or odd, absolute values, norms, fields, and Banach spaces have a quality which is either archimedean or non-archimedean.

Assumption 2 (Roots of Unity)**.** *In our approach to Pontryagin duality, a unitary character $G \to K$ of a locally compact abelian group G is an expression of the form:*

$$g \in G \mapsto e^{2\pi i \langle \gamma, g \rangle} \in K \tag{2.7}$$

where γ is an element of the Pontryagin dual $\hat{G}$ of G and $\langle \cdot, \cdot \rangle : \hat{G} \times G \to \mathbb{R}/\mathbb{Z}$ is the duality bracket/duality pairing (a continuous, $\mathbb{Z}$-bilinear homomorphism of locally compact abelian groups). To that end, ***we fix once and for all a choice of an embedding of the field of algebraic numbers $\overline{\mathbb{Q}}$ in*** *K, so that for any integer $d \geq 1$, we can write $e^{2\pi i/d}$ to denote our favorite primitive dth root of unity in K, and then exploit the familiar algebraic properties of complex exponential notation, such as:*

$$e^{2\pi i k/d} = \left(e^{2\pi i/d}\right)^k \tag{2.8}$$

and so that, for any prime p and any $n \geq 0$:

$$\left(e^{2\pi i/p^{n+1}}\right)^{p}=e^{2\pi i/p^{n}} \tag{2.9}$$

In this notation, every K-valued unitary character on $\mathbb{Z}_p$ is expressible as a map of the form:

$$\mathbb{Z}_p \to K \tag{2.10}$$

$$\mathfrak{z} \mapsto e^{2\pi i\{t\mathfrak{z}\}_p} \tag{2.11}$$

for some fixed $t \in \hat{\mathbb{Z}}_p$; note that multiplication of t and $\mathfrak{z}$ makes sense here, because both quantities lie in $\mathbb{Q}_p$.

In terms of our abuse-of-notation, in having identified $\hat{\mathbb{Z}}_p$ with $\mathbb{Z}[1/p]/\mathbb{Z}$, for any given $\mathfrak{y} \in \mathbb{Q}_p$, the fractional part of $\mathfrak{y}$ will be uniquely expressible as an irreducible p-power fraction in $[0,1)$ of the form k/p^n for some integers $n \geq 0$ and $k \in \{0,\ldots,p^n-1\}$. As such, we write $e^{2\pi i\{\mathfrak{y}\}_p}$ to denote the kth power of the primitive p^nth root of unity chosen along with our embedding $\overline{\mathbb{Q}} \hookrightarrow K$. Likewise, $e^{-2\pi i\{\mathfrak{y}\}_p}$ is the reciprocal of the root of unity denoted by $e^{2\pi i\{\mathfrak{y}\}_p}$.

2.1. **An Outline of this Section.** (p,q)-adic Fourier analysis is quite strange compared to classical Fourier analysis. The first word in this strangeness is the following result, originally proven by Schikhof as part of his 1967 PhD Dissertation [20]. (The monicker "The Fundamental Theorem of (p,q)-Adic Analysis" is our own.)

Theorem 4 (**The Fundamental Theorem of (p,q)-Adic Analysis**). *Let $q < \infty$. Then, for a function $\chi : \mathbb{Z}_p \to K$, the following are equivalent:*

I. χ is continuous.

II. χ is integrable with respect to the q-adic Haar probability measure on $\mathbb{Z}_p$.

III. The Fourier transform of χ exists for at least one $t \in \hat{\mathbb{Z}}_p$.

IV. The Fourier transform of χ exists for all $t \in \hat{\mathbb{Z}}_p$.

V. χ has a Fourier series representation which converges in K to $\chi(\mathfrak{z})$ uniformly with respect to $\mathfrak{z} \in \mathbb{Z}_p$.

It is because of this result that Schikhof and those who have come after him have dismissed (p,q)-adic analysis as "uninteresting" [19]. Without anything else to go on, it is very natural to make that conclusion, given the absurd rigidity asserted by the **Fundamental Theorem**. In Part III of this series of papers, we will explore how the **Fundamental Theorem** is *not*, in fact, the last word in (p,q)-adic analysis. However, for now, we hope the reader will be satisfied with an exposition of the basic tools of (p,q)-adic Fourier analysis and the surrounding constellation of ideas.

At a high-brow level, the theory Fourier analysis of functions taking values in non-archimedean fields emerges from the **Monna-Springer integral**, a more general construction used to formulate integration in abstract non-archimedean functional analysis; for details, see [11, 16], and also the appendices of [19]. Our exposition differs from these by virtue of its concreteness. In **Section 2.2**, we introduce the van der Put basis for $C(\mathbb{Z}_p, K)$, with which we can reduce the theory of (p,q)-adic functions to mere linear algebra. This has the double advantage of being both conceptually clear and very well-suited to direct computation and experimentation.

This concreteness hinges on an important but elementary observation: *by definition* an arbitrary field K contains an element denoted 0 (the additive identity) and an element denoted 1 (the multiplicative identity). As such the indicator function:

$$\left[\mathfrak{z} \overset{p^n}{\equiv} k\right] \tag{2.12}$$

makes sense as a map from $\mathbb{Z}_p$ to K. Moreover, because of $\mathbb{Z}_p$'s ultrametric topology, if K also happens to have a topology, this indicator function is then *guaranteed* to be an element of $C(\mathbb{Z}_p, K)$. With this insight and the almost trivial Fourier series representation of the indicator function, **Section 2.3** shows how, in conjunction with the van der Put Basis, we can define and compute the Fourier transform of any function in $C(\mathbb{Z}_p, K)$, provided K contains all p-power roots of unity. With this, we can easily prove the **Fundamental Theorem**.

Section 2.4 defines the (p,q)-adic Haar measure and uses it to construct the Fourier Transform in its usual integral operator form. We then define the Fourier-Stieltjes Transform of a (p,q)-adic measure and introduce convolutions. We then give the analogue for measures of the Fundamental Theorem of (p,q)-adic analysis. In **Section 2.5**, we introduce the p-adic Dirichlet kernel and use it to present our non-standard (though very natural) notion of the Radon-Nikodym derivative of a measure. Finally, in **Section 2.6**, we have proffered several exercises for the reader based on all this content. The solutions can be found at the end of the paper, in **Section 4**.

Note 1. We assume that the reader is familiar with Pontryagin duality and *real-valued* Haar measures on locally-compact abelian groups, particularly the p-adics. [8] is a standard references; [9, 14] cover the special cases relevant to number theorists (the p-adics, adèle rings, etc.).

2.2. **The van der Put Basis.** Kurt Mahler was the first to prove what is now a classic fact of p-adic analysis: every continuous function $\mathbb{Z}_p \to \mathbb{Q}_p$ can be uniquely expressed as a uniformly convergent series of polynomials:

$$f(\mathfrak{z}) = \sum_{n=0}^{\infty} a_n \binom{\mathfrak{z}}{n} \tag{2.13}$$

where:

$$\binom{\mathfrak{z}}{n} = \frac{1}{n!} \prod_{k=0}^{n-1} (\mathfrak{z} - k) \tag{2.14}$$

is the nth Binomial Coefficient polynomial. The set:

$$\left\{\binom{\mathfrak{z}}{n} : n \geq 0\right\} \tag{2.15}$$

is then called the **Mahler basis** of $C(\mathbb{Z}_p, \mathbb{Q}_p)$, and the ensuing series representation of f is known as f's **Mahler series** [15, 19].

However, Mahler series do not exist in (p,q) for the same reason that (p,q)-adic analytic functions fail to exist: there is no way to multiply a p-adic number by a q-adic number.

Fortunately for us, Marius van der Put discovered that an alternative family of functions that form a basis of $C(\mathbb{Z}_p, \mathbb{Q}_p)$; these are now named in his honor [15, 19].

Definition 4. The **p-adic van der Put basis** is the set:

$$\mathcal{B}_p \overset{\text{def}}{=} \left\{ \left[\mathfrak{z} \overset{p^{\lambda_p(n)}}{\equiv} n \right] : n \in \mathbb{N}_0 \right\} \tag{2.16}$$

Remark 2. Throughout, we will abbreviate "van der Put" as vdP.

Unlike the Mahler basis, the advantage of the vdP basis works for our (p,q)-adic functions in $C(\mathbb{Z}_p, K)$. In fact, it is the basis of the space of functions $\mathbb{Z}_p \to G$, where G is any abelian group!

Definition 5. Let G be an abelian group, written additively. For any function $f : \mathbb{Z}_p \to G$ and any $n \in \mathbb{N}_0$ the **nth van der Put (vdP) coefficient** of f is defined by:

$$c_n(f) \overset{\text{def}}{=} \begin{cases} f(0) & \text{if } n = 0 \\ f(n) - f(n_-) & \text{if } n \geq 1 \end{cases} \tag{2.17}$$

where n_- is the integer obtained by deleting the right-most digit in the p-adic expansion of n. That is, if:

$$n = d_0 + d_1 p + \cdots + d_{L-1} p^{L-1} + d_L p^L \tag{2.18}$$

then:

$$n_- = d_0 + d_1 p + \cdots + d_{L-1} p^{L-1} \tag{2.19}$$

This can be written more compactly as:

$$n_- = n - d_{\lambda_p(n)-1} p^{\lambda_p(n)-1} \tag{2.20}$$

because any $n \geq 0$ can be uniquely written as:

$$n = \sum_{\ell=0}^{\lambda_p(n)-1} d_\ell p^\ell \tag{2.21}$$

We then define $S\{f\}$ (also denoted $S_p\{f\}$; this latter notation is from the author's dissertation) as the formal sum:

$$S\{f\}(\mathfrak{z}) \overset{\text{def}}{=} \sum_{n=0}^{\infty} c_n(f) \left[\mathfrak{z} \overset{p^{\lambda_p(n)}}{\equiv} n \right] \tag{2.22}$$

Here, for each n:

$$\mathfrak{z} \mapsto c_n(f) \left[\mathfrak{z} \overset{p^{\lambda_p(n)}}{\equiv} n \right] \tag{2.23}$$

is the function from $\mathbb{Z}_p$ to G which outputs $c_n(f)$ if the congruence is satisfied and outputs 0 otherwise. We call $S\{f\}$ the **van der Put (vdP) series** of f. If G also has the structure of a metric space, we define $S\{f\}$'s **domain of convergence** as the set of all $\mathfrak{z} \in \mathbb{Z}_p$ so that $S\{f\}(\mathfrak{z})$ converges in G.

More generally, for any constants $c_n \in G$, by a **(formal) van der Put series**, we mean an expression of the form:

$$\sum_{n=0}^{\infty} c_n(f) \left[\mathfrak{z} \overset{p^{\lambda_p(n)}}{\equiv} n \right] \tag{2.24}$$

and, if G is a metric space, the domain of convergence of this series is then the set of $\mathfrak{z} \in \mathbb{Z}_p$ at which the series converges in G.

We begin our work with vdP series by showing that $\mathbb{N}_0$ is *always* in the domain of convergence of $S\{f\}$, and that $S\{f\}(n) = f(n)$ for all $n \in \mathbb{N}_0$. We do this in several steps, beginning with a simple observation about positive integers' p-adic digits.

Fact 1. *Let $k \in \mathbb{N}_1$. Then, a non-negative integer n has exactly k p-adic digits (i.e., $\lambda_p(n) = k$) if and only if $p^{k-1} \leq n \leq p^k - 1$.*

Using this, we can perform what the author calls a **λ-decomposition**. This is the technique of taking a sum over $\mathbb{N}_0$ and decomposing it into a sum over the subsets of $\mathbb{N}_0$ on which λ_p is constant.

Proposition 1. *Let G be an abelian group, and let $f : \mathbb{Z}_p \to G$. Then, for all $N \in \mathbb{N}_0$:*

$$\sum_{n=0}^{p^N-1} c_n(f)\left[\mathfrak{z} \overset{p^{\lambda_p(n)}}{\equiv} n\right] = f\left([\mathfrak{z}]_{p^N}\right), \ \forall \mathfrak{z} \in \mathbb{Z}_p \tag{2.25}$$

Remark 3. In his dissertation, the author called this the **(Nth) truncated van der Put identity**. We shall use that terminology here as well.

Proof: Fixing $\mathfrak{z} \in \mathbb{Z}_p$, we start by applying a λ-decomposition to f's vdP series:

$$S\{f\}(\mathfrak{z}) = \sum_{n=0}^{\infty} c_n(f)\left[\mathfrak{z} \overset{p^{\lambda_p(n)}}{\equiv} n\right] = c_0(f) + \sum_{k=1}^{\infty}\sum_{n=p^{k-1}}^{p^k-1} c_n(f)\left[\mathfrak{z} \overset{p^k}{\equiv} n\right] \tag{2.26}$$

Next, letting $k \geq 1$ be arbitrary, since n is being summed over all numbers with k p-adic digits, we can write:

$$\sum_{n=p^{k-1}}^{p^k-1} c_n(f)\left[\mathfrak{z} \overset{p^k}{\equiv} n\right] = \sum_{n=p^{k-1}}^{p^k-1} c_n(f)\left[\mathfrak{z} \overset{p^k}{\equiv} n\right][\lambda_p(n) = k] \tag{2.27}$$

Now, note that there is *at most* one value of $n \in \left\{p^{k-1}, \ldots, p^k - 1\right\}$ which is congruent to $\mathfrak{z}$ mod p^k: $n = [\mathfrak{z}]_{p^k}$. Because of this, the Iverson bracket kills all the ns in our sum except $n = [\mathfrak{z}]_{p^k}$, leaving us with:

$$\sum_{n=p^{k-1}}^{p^k-1} c_n(f)\left[\mathfrak{z} \overset{p^k}{\equiv} n\right] = c_{[\mathfrak{z}]_{p^k}}(f)\left[\mathfrak{z} \overset{p^k}{\equiv} [\mathfrak{z}]_{p^k}\right]\left[\lambda_p\left([\mathfrak{z}]_{p^k}\right) = k\right] \tag{2.28}$$

Since $\mathfrak{z}$ is always congruent to $[\mathfrak{z}]_{p^k}$ mod p^k, the Iverson bracket $\left[\mathfrak{z} \overset{p^k}{\equiv} [\mathfrak{z}]_{p^k}\right]$ evaluates to 1 for all $\mathfrak{z} \in \mathbb{Z}_p$. This gives:

$$\sum_{n=p^{k-1}}^{p^k-1} c_n(f)\left[\mathfrak{z} \overset{p^k}{\equiv} n\right] = c_{[\mathfrak{z}]_{p^k}}(f)\left[\lambda_p\left([\mathfrak{z}]_{p^k}\right) = k\right] \tag{2.29}$$

The Iverson bracket on the right-hand side of (2.29) tells us that $[\mathfrak{z}]_{p^k}$ must have k p-adic digits. As such, writing $\mathfrak{z}$ in Hensel series form as:

$$\mathfrak{z} = \sum_{j=0}^{\infty} d_j p^j \tag{2.30}$$

we see that:

$$[\mathfrak{z}]_{p^k} = \sum_{j=0}^{k-1} d_j p^j \tag{2.31}$$

where $d_{k-1} \neq 0$. Thus, applying the subscript $-$ operator from the definition of the vdP coefficients, we get:

$$\left([\mathfrak{z}]_{p^k}\right)_- = [\mathfrak{z}]_{p^k} - d_{k-1}p^{k-1} = \sum_{j=0}^{k-2} d_j p^j = [\mathfrak{z}]_{p^{k-1}} \tag{2.32}$$

Since:

$$c_{[\mathfrak{z}]_{p^k}}(f) = f\left([\mathfrak{z}]_{p^k}\right) - f\left(\left([\mathfrak{z}]_{p^k}\right)_-\right) = f\left([\mathfrak{z}]_{p^k}\right) - f\left([\mathfrak{z}]_{p^{k-1}}\right) \tag{2.33}$$

we then have:

$$c_{[\mathfrak{z}]_{p^k}}(f)\left[\lambda_p\left([\mathfrak{z}]_{p^k}\right) = k\right] \overset{K}{=} \left(f\left([\mathfrak{z}]_{p^k}\right) - f\left([\mathfrak{z}]_{p^{k-1}}\right)\right)\left[\lambda_p\left([\mathfrak{z}]_{p^k}\right) = k\right] \tag{2.34}$$

Claim 1. The Iverson bracket on the right-hand side of (2.34) can be removed.

Proof of claim: If $\lambda_p\left([\mathfrak{z}]_{p^k}\right) = k$, then the Iverson bracket is 1, and it disappears on its own without causing us any trouble. So, suppose $\lambda_p\left([\mathfrak{z}]_{p^k}\right) \neq k$. Then, d_{k-1}, the right-most digit of $[\mathfrak{z}]_{p^k}$, is 0. But then, $[\mathfrak{z}]_{p^k} = [\mathfrak{z}]_{p^{k-1}}$, which makes the right-hand side of (2.34) vanish because $f\left([\mathfrak{z}]_{p^k}\right) - f\left([\mathfrak{z}]_{p^{k-1}}\right)$ is then equal to 0. So, *any case* which would cause the Iverson bracket to vanish causes $f\left([\mathfrak{z}]_{p^k}\right) - f\left([\mathfrak{z}]_{p^{k-1}}\right)$ to vanish. This tells us that the Iverson bracket in (2.34) isn't actually needed. As such, we are justified in dropping the Iverson bracket altogether:

$$c_{[\mathfrak{z}]_{p^k}}(f)\left[\lambda_p\left([\mathfrak{z}]_{p^k}\right) = k\right] = f\left([\mathfrak{z}]_{p^k}\right) - f\left([\mathfrak{z}]_{p^{k-1}}\right) \tag{2.35}$$

This proves the claim.

Finally, we write:

$$\begin{aligned}
\sum_{n=0}^{p^N-1} c_n(f)\left[\mathfrak{z} \overset{p^{\lambda_p(n)}}{\equiv} n\right] &= c_0(f) + \sum_{k=1}^{N}\sum_{n=p^{k-1}}^{p^k-1} c_n(f)\left[\mathfrak{z} \overset{p^k}{\equiv} n\right] \\
&= c_0(f) + \sum_{k=1}^{N} c_{[\mathfrak{z}]_{p^k}}(f)\left[\lambda_p\left([\mathfrak{z}]_{p^k}\right) = k\right] \\
&= c_0(f) + \underbrace{\sum_{k=1}^{N}\left(f\left([\mathfrak{z}]_{p^k}\right) - f\left([\mathfrak{z}]_{p^{k-1}}\right)\right)}_{\text{telescoping}} \\
\left(c_0(f) = f\left([\mathfrak{z}]_{p^0}\right) = f(0)\right); &= f(0) + f\left([\mathfrak{z}]_{p^N}\right) - f(0) \\
&= f\left([\mathfrak{z}]_{p^N}\right)
\end{aligned}$$

and we are done.

Q.E.D.

Lemma 1 (***van der Put Limit***). *Let G be an abelian group which is also a complete metric space, let $f:\mathbb{Z}_p\rightarrow G$, and let D be the domain of convergence of $S\{f\}$. Then:*

$$S\{f\}(\mathfrak{z})\overset{G}{=}\lim_{k\rightarrow\infty}f\left([\mathfrak{z}]_{p^k}\right),\ \forall\mathfrak{z}\in D \tag{2.36}$$

where, as indicated, the limit converges in G. In particular, we have that $\mathbb{N}_0\subseteq D$, and that $S\{f\}(\mathfrak{z})$ converges to $f(\mathfrak{z})$ in the discrete topology for all $\mathfrak{z}\in\mathbb{N}_0$.

Remark 4. This result is part of Exercise 62.B on page 192 of [19].

Proof: Using the **van der Put identity**:

$$\sum_{n=0}^{p^N-1}c_n(f)\left[\mathfrak{z}\overset{p^{\lambda_p(n)}}{\equiv}n\right]=f\left([\mathfrak{z}]_{p^N}\right) \tag{2.37}$$

So, let $\mathfrak{z}\in\mathbb{N}_0$. Then, $[\mathfrak{z}]_{p^N}=\mathfrak{z}$ for all $N\geq\lambda_p(\mathfrak{z})$, and so:

$$N\geq\lambda_p(\mathfrak{z})\Rightarrow\sum_{n=0}^{p^N-1}c_n(f)\left[\mathfrak{z}\overset{p^{\lambda_p(n)}}{\equiv}n\right]=f(\mathfrak{z}) \tag{2.38}$$

Thus, the partial sums of $S\{f\}(\mathfrak{z})$ converge to $f(\mathfrak{z})$ in the discrete topology when $\mathfrak{z}\in\mathbb{N}_0$. For all $\mathfrak{z}\in\mathbb{Z}_p^{\prime}$, meanwhile, the partial sums of $S\{f\}(\mathfrak{z})$ converge if and only if $\lim_{k\rightarrow\infty}f\left([\mathfrak{z}]_{p^k}\right)$ converges in G, in which case, the vdP identity tells us that:

$$S\{f\}(\mathfrak{z})\overset{G}{=}\lim_{k\rightarrow\infty}f\left([\mathfrak{z}]_{p^k}\right),\ \forall\mathfrak{z}\in D \tag{2.39}$$

Q.E.D.

With this, we can now show that the vdP series represents all continuous functions $\mathbb{Z}_p\rightarrow\mathbb{F}$, where $\mathbb{F}$ is a metrically complete valued field.

Theorem 5. *Let $\mathbb{F}$ be any metrically complete valued field with absolute value $|\cdot|_q$.*

*I. Every continuous $f:\mathbb{Z}_p\rightarrow\mathbb{F}$ admits a unique representation as a **van der Put (vdP) series**:*

$$f(\mathfrak{z})\overset{\mathbb{F}}{=}\sum_{n=0}^{\infty}c_n(f)\left[\mathfrak{z}\overset{p^{\lambda_p(n)}}{\equiv}n\right],\ \forall\mathfrak{z}\in\mathbb{Z}_p \tag{2.40}$$

This series converges to f uniformly with respect to $\mathfrak{z}\in\mathbb{Z}_p$.

II. If $\mathbb{F}$ is non-archimedean, then $f:\mathbb{Z}_p\rightarrow\mathbb{F}$ is continuous if and only if $|c_n(f)|_q$ tends to 0 in $\mathbb{R}$ as $n\rightarrow\infty$.

Proof:
I. First, observe that given any vdP series, the series' partial sums are locally constant, and are therefore continuous. Thus, if a vdP series converges uniformly, its sequence of partial sums is a uniformly convergent sequence of continuous functions, which by elementary analysis necessarily converges to a continuous limit. So, every uniformly convergent vdP series sums to a continuous function.

To finish, we just need to show that every continuous function can be represented as a uniformly convergent vdP series. So, letting f be continuous and fixing $\mathfrak{z} \in \mathbb{Z}_p$, since $[\mathfrak{z}]_{p^n}$ converges in $\mathbb{Z}_p$ to $\mathfrak{z}$ as $n \to \infty$, we have that:

$$\lim_{n\to\infty} f\left([\mathfrak{z}]_{p^n}\right) \overset{\mathbb{F}}{=} f(\mathfrak{z}), \ \forall \mathfrak{z} \in \mathbb{Z}_p \tag{2.41}$$

(Note: so far, the convergence is merely point-wise.) Thus, by the **van der Put identity**, since:

$$S\{f\}(\mathfrak{z}) \overset{\mathbb{F}}{=} \lim_{n\to\infty} f\left([\mathfrak{z}]_{p^n}\right) \tag{2.42}$$

for all $\mathfrak{z}$ for which the limit exists, we have:

$$S\{f\}(\mathfrak{z}) \overset{\mathbb{F}}{=} \lim_{n\to\infty} f\left([\mathfrak{z}]_{p^n}\right) \overset{\mathbb{F}}{=} f(\mathfrak{z}), \ \forall \mathfrak{z} \in \mathbb{Z}_p \tag{2.43}$$

Thus, $S\{f\}$ represents f point-wise. To see that this convergence is *uniform*, note that since $\mathbb{Z}_p$ is compact, $f(\mathfrak{z})$'s continuity on $\mathbb{Z}_p$ is uniform. So, letting $\epsilon > 0$, choose $\delta > 0$ so that:

$$\mathfrak{x}, \mathfrak{y} \in \mathbb{Z}_p \ \& \ |\mathfrak{x} - \mathfrak{y}|_p < \delta \Rightarrow |f(\mathfrak{x}) - f(\mathfrak{y})|_q$$

where, as usual, $|\cdot|_q$ is the absolute value on $\mathbb{F}$. Since $\mathfrak{z}$ is always congruent to $[\mathfrak{z}]_{p^n}$ mod p^n, we have:

$$\left|\mathfrak{z} - [\mathfrak{z}]_{p^n}\right|_p \leq p^{-n}, \ \forall \mathfrak{z} \in \mathbb{Z}_p, \ \forall n \in \mathbb{N}_0$$

Thus, choosing N_δ so that $n \geq N_\delta$ implies $p^{-n} < \delta$, we have:

$$n \geq N_\delta \Rightarrow \left|\mathfrak{z} - [\mathfrak{z}]_{p^n}\right|_p \leq p^{-n} \ \forall \mathfrak{z} \in \mathbb{Z}_p \Rightarrow \left|f(\mathfrak{z}) - f\left([\mathfrak{z}]_{p^n}\right)\right|_q < \epsilon \ \forall \mathfrak{z} \in \mathbb{Z}_p$$

and so, $n \geq N_\delta$ implies:

$$\sup_{\mathfrak{z}\in\mathbb{Z}_p} \left|f(\mathfrak{z}) - f\left([\mathfrak{z}]_{p^n}\right)\right|_q < \epsilon$$

which proves that the $f\left([\mathfrak{z}]_{p^n}\right)$s converge to $f(\mathfrak{z})$ uniformly over $\mathbb{Z}_p$, which—by (2.43)—then proves the uniform convergence of $S\{f\}$ to f.

II. If $\mathbb{F}$ is non-archimedean, the convergence of $S\{f\}$ is then equivalent to the decay of $|c_n(f)|_q$ in $\mathbb{R}$ to 0 as $n \to \infty$.

Q.E.D.

2.3. **Fourier Series.** An alternative way to arrive at the indicator functions for compact-open subsets of $\mathbb{Z}_p$ is through the notion of locally constant functions.

Definition 6. Let $\mathbb{F}$ be a field and let $N \in \mathbb{N}_0$. We say a function $f : \mathbb{Z}_p \to \mathbb{F}$ is **locally constant mod** p^N if:

$$f(\mathfrak{z}) = f\left([\mathfrak{z}]_{p^N}\right), \ \forall \mathfrak{z} \in \mathbb{Z}_p \tag{2.44}$$

That is, f's value at $\mathfrak{z}$ depends only on the value of $\mathfrak{z}$ mod p^N. We say a function $f : \mathbb{Z}_p \to \mathbb{F}$ is **locally constant** if it is locally constant mod p^N for some $N \in \mathbb{N}_0$.

We write $\mathcal{S}(\mathbb{Z}_p, \mathbb{F})$ to denote **the set of all locally constant functions** $f : \mathbb{Z}_p \to \mathbb{F}$. This is a vector space over $\mathbb{F}$, under the operations of scalar multiplication and point-wise additions. $\mathcal{S}(\mathbb{Z}_p, \mathbb{F})$ also becomes an $\mathbb{F}$-algebra if we equip it with point-wise multiplication of functions.

Remark 5. The set $\mathcal{S}(\mathbb{Z}_p,\mathbb{C})$ is known as the set of **Schwartz-Bruhat functions** on $\mathbb{Z}_p$, viz. [26]. Consequently we shall also call $\mathcal{S}(\mathbb{Z}_p,\mathbb{F})$ the set of $\mathbb{F}$-valued Schwartz-Bruhat functions on $\mathbb{Z}_p$.

Remark 6. The indicator function:

$$\left[\mathfrak{z}\overset{p^n}{\equiv}k\right] \tag{2.45}$$

for the compact-open set $k+p^n\mathbb{Z}_p$ is locally constant mod p^n. Moreover, every element of $\mathcal{S}(\mathbb{Z}_p,\mathbb{F})$ is of the form:

$$\sum_{j=0}^{J}a_j\left[\mathfrak{z}\overset{p^{n_j}}{\equiv}k_j\right] \tag{2.46}$$

for $J\in\mathbb{N}_0$, $a_0,\ldots,a_J\in\mathbb{F}$, $n_0,\ldots,n_J\in\mathbb{N}_0$, and $k_j\in\{0,\ldots,p^{n_j}-1\}$for all j. Moreover, observe that every locally constant function mod p^M can be uniquely written as a linear combination of finitely many indicator functions:

$$f(\mathfrak{z})=\sum_{n=0}^{p^N-1}f(n)\left[\mathfrak{z}\overset{p^N}{\equiv}n\right] \tag{2.47}$$

where N is any integer $\geq M$.

Consequently, as an algebra, $\mathcal{S}(\mathbb{Z}_p,\mathbb{F})$ is generated by the indicator functions of the compact-opens.

Proposition 2. *Let $\mathbb{F}$ be any topological field, and let $f:\mathbb{Z}_p\rightarrow\mathbb{F}$ be locally constant. Then, f is continuous.*

Proof: It's a simple exercise.
Q.E.D.

Proposition 3. *Let $\mathbb{F}$ be any metrically complete valued field, and let $f\in C(\mathbb{Z}_p,\mathbb{F})$. Then, $f\left([\mathfrak{z}]_{p^n}\right)$ converges to $f(\mathfrak{z})$ uniformly with respect to $\mathfrak{z}\in\mathbb{Z}_p$ as $n\rightarrow\infty$.*

Proof: We proved this as part of **Theorem 5**.
Q.E.D.

Note 2. Having covered the generalities, from here on out, UNLESS STATED OTHERWISE, ALL FUNCTIONS TAKE VALUES IN OUR q-ADIC FIELD K.

Now, we introduce *the* most important formula in (p,q)-adic analysis.

Proposition 4 (**Fourier series of** $\left[\mathfrak{z}\overset{p^n}{\equiv}k\right]$)**.** *For all $n\in\mathbb{N}_0$ and $k\in\mathbb{Z}$:*

$$\left[\mathfrak{z}\overset{p^n}{\equiv}k\right]=\frac{1}{p^n}\sum_{|t|_p\leq p^n}e^{2\pi i\{t(\mathfrak{z}-k)\}_p},\ \forall\mathfrak{z}\in\mathbb{Z}_p \tag{2.48}$$

Remark 7. We can also write (2.48) as:

$$\left[\mathfrak{z}\overset{p^n}{\equiv}k\right]=\frac{1}{p^n}\sum_{j=0}^{p^n-1}e^{2\pi i\left\{j\frac{\mathfrak{z}-k}{p^n}\right\}_p},\ \forall\mathfrak{z}\in\mathbb{Z}_p \tag{2.49}$$

Proof: Sum the finite geometric series:

$$\sum_{|t|_p \leq p^n} e^{2\pi i \{t(\mathfrak{z}-k)\}_p} = \sum_{j=0}^{p^n-1} e^{2\pi i \left\{j \frac{\mathfrak{z}-k}{p^n}\right\}_p} = \sum_{j=0}^{p^n-1} \left(\exp\left(2\pi i \left\{\frac{\mathfrak{z}-k}{p^n}\right\}_p\right)\right)^j \tag{2.50}$$

If K is non-archimedean, the stipulation that K's residue field has characteristic $\neq p$ then forces $\mathrm{char}K \neq p$, so we can divide by p^n and get the result we want.

Q.E.D.

With this formula, we can directly compute the Fourier series representation of a continuous (p,q)-adic function by writing out the function's vdP series, applying **Proposition 4** to the indicator functions of the vdP basis, and then interchanging sums.

Theorem 6. *Let $f \in C(\mathbb{Z}_p, K)$. Then, the function $\hat{f} : \hat{\mathbb{Z}}_p \to K$ defined by:*

$$\hat{f}(t) \overset{K}{=} \sum_{n=\frac{1}{p}|t|_p}^{\infty} \frac{c_n(f)}{p^{\lambda_p(n)}} e^{-2\pi i n t} \tag{2.51}$$

converges uniformly with respect to $t \in \hat{\mathbb{Z}}_p$. $f(\mathfrak{z})$ then has the Fourier series representation:

$$f(\mathfrak{z}) \overset{K}{=} \sum_{t \in \hat{\mathbb{Z}}_p} \hat{f}(t) e^{2\pi i \{t\mathfrak{z}\}_p} \tag{2.52}$$

This series converges uniformly with respect to $\mathfrak{z} \in \mathbb{Z}_p$. Moreover, this is the unique *Fourier series which uniformly converges to f in K.*

Proof: We use the vdP series, first pulling out the $n = 0$ term and then re-writing the indicator functions as Fourier series. We then split the t sum into sums of level sets of constant p-adic absolute value. This gives us:

$$\begin{aligned}
f(\mathfrak{z}) &\overset{K}{=} \sum_{n=0}^{\infty} c_n(f) \left[\mathfrak{z} \overset{p^{\lambda_p(n)}}{\equiv} n\right] \\
&\overset{K}{=} c_0(f) + \sum_{n=1}^{\infty} c_n(f) \left(\frac{1}{p^{\lambda_p(n)}} \sum_{|t|_p \leq p^{\lambda_p(n)}} e^{2\pi i \{t(\mathfrak{z}-n)\}_p}\right) \\
&\overset{K}{=} c_0(f) + \sum_{n=1}^{\infty} \frac{c_n(f)}{p^{\lambda_p(n)}} \left(1 + \sum_{k=1}^{\lambda_p(n)} \sum_{|t|_p = p^k} e^{2\pi i \{t(\mathfrak{z}-n)\}_p}\right) \\
&\overset{K}{=} c_0(f) + \sum_{n=1}^{\infty} \frac{c_n(f)}{p^{\lambda_p(n)}} + \sum_{n=1}^{\infty} \sum_{k=1}^{\lambda_p(n)} \sum_{|t|_p = p^k} \frac{c_n(f)}{p^{\lambda_p(n)}} e^{2\pi i \{t(\mathfrak{z}-n)\}_p}
\end{aligned}$$

Next, we recall our observation that for $n, k \in \mathbb{N}_1$, the equality $\lambda_p(n) = k$ occurs if and only if $p^{k-1} \leq n \leq p^k - 1$. This lets us re-index our double sum like so:

$$\sum_{n=1}^{\infty} \sum_{k=1}^{\lambda_p(n)} = \sum_{k=1}^{\infty} \sum_{n=p^{k-1}}^{\infty} \tag{2.53}$$

Thus, our triple sum becomes:

$$\sum_{n=1}^{\infty}\sum_{k=1}^{\lambda_p(n)}\sum_{\left|t\right|_p=p^k}=\sum_{k=1}^{\infty}\sum_{n=p^{k-1}}^{\infty}\sum_{\left|t\right|_p=p^k} \tag{2.54}$$

and so:

$$\begin{aligned} f\left(\mathfrak{z}\right) &\overset{K}{=} c_0\left(f\right)+\sum_{n=1}^{\infty}\frac{c_n\left(f\right)}{p^{\lambda_p(n)}}+\sum_{k=1}^{\infty}\sum_{n=p^{k-1}}^{\infty}\sum_{\left|t\right|_p=p^k}\frac{c_n\left(f\right)}{p^{\lambda_p(n)}}e^{2\pi i\left\{t\left(\mathfrak{z}-n\right)\right\}_p} \\ &\overset{!}{=} c_0\left(f\right)+\sum_{n=1}^{\infty}\frac{c_n\left(f\right)}{p^{\lambda_p(n)}}+\sum_{k=1}^{\infty}\sum_{\left|t\right|_p=p^k}\left(\sum_{n=p^{k-1}}^{\infty}\frac{c_n\left(f\right)}{p^{\lambda_p(n)}}e^{-2\pi int}\right)e^{2\pi i\left\{t\mathfrak{z}\right\}_p}\end{aligned}$$

As for the interchange of infinite sums that occurs in the step marked (!), **Theorem 5** tells us that f's continuity makes f's vdP series uniformly convergent with respect to $\mathfrak{z}$, which justifies our interchange of infinite sums.

Next, note that when $\left|t\right|_p=p^k$, k is then equal to $-v_p\left(t\right)$, and so:

$$\sum_{\left|t\right|_p=p^k}\sum_{n=p^{k-1}}^{\infty}=\sum_{\left|t\right|_p=p^k}\sum_{n=p^{-v_p(t)-1}}^{\infty}=\sum_{\left|t\right|_p=p^k}\sum_{n=\frac{1}{p}\left|t\right|_p}^{\infty} \tag{2.55}$$

By writing:

$$\sum_{k=1}^{\infty}\sum_{\left|t\right|_p=p^k}=\sum_{t\in\hat{\mathbb{Z}}_p\backslash\{0\}} \tag{2.56}$$

we get:

$$\begin{aligned} f\left(\mathfrak{z}\right) &\overset{K}{=} c_0\left(f\right)+\sum_{n=1}^{\infty}\frac{c_n\left(f\right)}{p^{\lambda_p(n)}}+\sum_{k=1}^{\infty}\sum_{\left|t\right|_p=p^k}\left(\sum_{n=\frac{1}{p}\left|t\right|_p}^{\infty}\frac{c_n\left(f\right)}{p^{\lambda_p(n)}}e^{-2\pi int}\right)e^{2\pi i\left\{t\mathfrak{z}\right\}_p} \\ &= \sum_{n=0}^{\infty}\frac{c_n\left(f\right)}{p^{\lambda_p(n)}}+\sum_{t\in\hat{\mathbb{Z}}_p\backslash\{0\}}\left(\sum_{n=\frac{1}{p}\left|t\right|_p}^{\infty}\frac{c_n\left(f\right)}{p^{\lambda_p(n)}}e^{-2\pi int}\right)e^{2\pi i\left\{t\mathfrak{z}\right\}_p}\end{aligned}$$

Recognizing the n-sum on the left as the $t=0$ case of the sum on the right, we get:

$$f\left(\mathfrak{z}\right)\overset{K}{=}\sum_{t\in\hat{\mathbb{Z}}_p}\left(\sum_{n=\frac{1}{p}\left|t\right|_p}^{\infty}\frac{c_n\left(f\right)}{p^{\lambda_p(n)}}e^{-2\pi int}\right)e^{2\pi i\left\{t\mathfrak{z}\right\}_p} \tag{2.57}$$

Since we started with a series that converged in K uniformly with respect to $\mathfrak{z}\in\mathbb{Z}_p$, this new formula of ours is also uniformly convergent with respect to those $\mathfrak{z}$. Furthermore, since all the steps are reversible, this also proves the uniqueness of f's Fourier series representation.

As for the matter of convergence, there are two cases, depending on the quality of K.

If K is <u>non-archimedean</u>, we note that since the characteristic of K's residue field is $\neq p$, the number p has an q-adic absolute value of 1, as do the roots of unity $e^{-2\pi int}$. Because of this:

$$\left|\frac{c_n\left(f\right)}{p^{\lambda_p(n)}}e^{-2\pi int}\right|_q=\left|c_n\left(f\right)\right|_q \tag{2.58}$$

Since f is continuous, the right-hand side tends to 0 as $n \to \infty$. Since K is non-archimedean, this decay then guarantees the series defining $\hat{f}(t)$ will converge in K. Moreover, this convergence is uniform with respect to $t \in \hat{\mathbb{Z}}_p$, since the q-adic decay of f's vdP coefficients occurs independently of t.

On the other hand, suppose K is archimedean. Then, since f is continuous, it is integrable over $\mathbb{Z}_p$ with respect to the *real-valued* Haar probability measure on $\mathbb{Z}_p$, denoted $d\mathfrak{z}$. Since this measure is a continuous linear functional, the uniform convergence of f's vdP series lets us compute the integral of f term by term:

$$\int_{\mathbb{Z}_p} f(\mathfrak{z})\, d\mathfrak{z} \overset{K}{=} \sum_{n=0}^{\infty} c_n(f) \int_{\mathbb{Z}_p} \left[\mathfrak{z} \overset{p^{\lambda_p(n)}}{\equiv} n\right] d\mathfrak{z} \overset{K}{=} \sum_{n=0}^{\infty} \frac{c_n(f)}{p^{\lambda_p(n)}} \tag{2.59}$$

Performing a λ-decomposition on the series on the right yields:

$$\sum_{n=0}^{\infty} \frac{c_n(f)}{p^{\lambda_p(n)}} \overset{K}{=} c_0(f) + \sum_{k=1}^{\infty} \frac{1}{p^k} \sum_{n=p^{k-1}}^{p^k-1} c_n(f) \tag{2.60}$$

Since this converges, the root test for series convergence forces:

$$\limsup_{k\to\infty} \left(\frac{1}{p}\left|\sum_{n=p^{k-1}}^{p^k-1} c_n(f)\right|_{\infty}^{1/k}\right) \leq 1 \tag{2.61}$$

Next, we make the estimate:

$$\left|\sum_{n=p^{k-1}}^{p^k-1} c_n(f)\right|_{\infty} \leq \sum_{n=p^{k-1}}^{p^k-1} \|f\|_{p,\infty} = (p-1)\,p^{k-1}\,\|f\|_{p,\infty} \tag{2.62}$$

so:

$$\frac{1}{p}\left|\sum_{n=p^{k-1}}^{p^k-1} c_n(f)\right|^{1/k} \leq \left(\frac{p-1}{p}\|f\|_{p,\infty}\right)^{1/k} \tag{2.63}$$

Now, suppose the inequality in the root test is an equality. Then:

$$1 = \limsup_{k\to\infty} \frac{1}{p}\left|\sum_{n=p^{k-1}}^{p^k-1} c_n(f)\right|^{1/k} \leq \limsup_{k\to\infty} \left(\frac{p-1}{p}\|f\|_{p,\infty}\right)^{1/k} \tag{2.64}$$

By definition, this means that there are infinitely many k for which:

$$\left(\frac{p-1}{p}\|f\|_{p,\infty}\right)^{1/k} \geq 1 \tag{2.65}$$

This then forces:

$$\|f\|_{p,\infty} \geq \frac{p}{p-1} \tag{2.66}$$

Since f is continuous on the compact set $\mathbb{Z}_p$, f is bounded. Thus, by dividing f by a sufficiently large positive real number, we can ensure that $\|f\|_{p,\infty} < \frac{p}{p-1}$, which then forces the root test's inequality to be strict, which then guarantees the *absolute* convergence of the series:

$$\hat{f}(0) = \sum_{n=0}^{\infty} \frac{c_n(f)}{p^{\lambda_p(n)}} \tag{2.67}$$

Consequently, the upper bound in the following triangle inequality estimate:

$$\left|\hat{f}(t)\right|_{\infty} \leq \sum_{n=\frac{1}{p}|t|_{p}}^{\infty} \frac{|c_{n}(f)|_{\infty}}{p^{\lambda_{p}(n)}} \leq \sum_{n=0}^{\infty} \frac{|c_{n}(f)|_{\infty}}{p^{\lambda_{p}(n)}},\ \forall t \in \hat{\mathbb{Z}}_{p} \tag{2.68}$$

converges, and so $\hat{f}$ converges uniformly, by the M-test.

Q.E.D.

2.4. Measures and the Fourier-Stieltjes Transform.

Assumption 3. *THROUGHOUT THIS SUBSECTION, UNLESS STATED OTHERWISE, WE ASSUME K IS NON-ARCHIMEDEAN.*

Picking up where we left off in the Preliminaries of **Section 1**, there are several more definitions we need in order to take things further.

Definition 7. As in classical analysis, we can multiply a measure by a continuous function to get a new measure. For any $g \in C(\mathbb{Z}_{p}, K)$, we define $g(\mathfrak{z})\,d\mu(\mathfrak{z})$ as the measure which sends a given $f \in C(\mathbb{Z}_{p}, K)$ to the image of $f(\mathfrak{z})\,g(\mathfrak{z})$ under $d\mu$:

$$\int_{\mathbb{Z}_{p}} f(\mathfrak{z})\left(g(\mathfrak{z})\,d\mu(\mathfrak{z})\right) \overset{\text{def}}{=} \int_{\mathbb{Z}_{p}} \left(f(\mathfrak{z})\,g(\mathfrak{z})\right) d\mu(\mathfrak{z}) \tag{2.69}$$

That is to say, $C(\mathbb{Z}_{p}, K)'$ **has the structure of a** $C(\mathbb{Z}_{p}, K)$**-module**. In particular, the bilinear map:

$$C(\mathbb{Z}_{p}, K) \times C(\mathbb{Z}_{p}, K)' \to C(\mathbb{Z}_{p}, K)' \tag{2.70}$$

defined by:

$$(g, d\mu) \mapsto g(\mathfrak{z})\,d\mu(\mathfrak{z}) \tag{2.71}$$

is continuous, with:

$$\left\|g(\mathfrak{z})\,d\mu(\mathfrak{z})\right\| \leq \|g\|_{p,q}\,\|d\mu\| \tag{2.72}$$

Next, we say $d\mu \in C(\mathbb{Z}_{p}, K)'$ is **translation-invariant** whenever:

$$\int_{\mathbb{Z}_{p}} f(\mathfrak{z}+\mathfrak{a})\,d\mu(\mathfrak{z}) = \int_{\mathbb{Z}_{p}} f(\mathfrak{z})\,d\mu(\mathfrak{z}),\ \forall \mathfrak{a} \in \mathbb{Z}_{p},\ \forall f \in C(\mathbb{Z}_{p}, K) \tag{2.73}$$

Additionally, given $U \subseteq \mathbb{Z}_{p}$, if the function $[\mathfrak{z} \in U]$ is continuous, we define $\mu(U)$ (the **μ-measure** of U) as the image of $[\mathfrak{z} \in U]$ under $d\mu$. In integral form, this can be written as:

$$\int_{U} d\mu(\mathfrak{z}) \tag{2.74}$$

We begin with the **triangle inequality for integrals**:

Proposition 5. *Let $d\mu \in C(\mathbb{Z}_{p}, K)'$. Then:*

$$\left|\int_{\mathbb{Z}_{p}} f(\mathfrak{z})\,d\mu(\mathfrak{z})\right|_{q} \leq \|f\|_{p,q}\,\|d\mu\|,\ \forall f \in C(\mathbb{Z}_{p}, K) \tag{2.75}$$

Proof: $d\mu$ is defined as a continuous linear functional, and $\|d\mu\|$ is its norm as a functional.

Q.E.D.

Next, we introduce the Haar measure.

Definition 8. The (p,q)-adic Haar measure, denoted $d\mathfrak{z}$, is the element of $C(\mathbb{Z}_p,K)'$ defined by the rule:

$$\int_{\mathfrak{a}+p^n\mathbb{Z}_p} d\mathfrak{z} \overset{\text{def}}{=} \int_{\mathbb{Z}_p} \left[\mathfrak{z} \overset{p^n}{\equiv} \mathfrak{a}\right] d\mathfrak{z} \overset{\text{def}}{=} \frac{1}{p^n} \tag{2.76}$$

for all $n \geq 0$ and all $\mathfrak{a} \in \mathbb{Z}_p$. Note that this does, in fact, define $d\mathfrak{z}$ as an element of $C(\mathbb{Z}_p,K)'$, thanks to the vdP basis. We then have that:

$$\int_{\mathbb{Z}_p} f(\mathfrak{z})\, d\mathfrak{z} \overset{K}{=} \int_{\mathbb{Z}_p} \left(\sum_{n=0}^{\infty} c_n(f)\left[\mathfrak{z} \overset{p^{\lambda_p(n)}}{\equiv} n\right]\right) d\mathfrak{z} \overset{K}{=} \sum_{n=0}^{\infty} \frac{c_n(f)}{p^{\lambda_p(n)}} \tag{2.77}$$

for all $f \in C(\mathbb{Z}_p,K)$. Since K is non-archimedean and q-adic, and $\operatorname{char}(K) \neq p$, and since the continuity of f guarantees that $|c_n(f)|_q \to 0$ as $n \to \infty$, the series on the far right converges in K.

Our next theorem establishes the usual characterization-definition of the Haar probability measure as the unique translation-invariant measure satisfying a certain normalization condition. Moreover, we show that this measure is what we get when we integrate (p,q)-adic functions by taking limits of their p-adic "Riemann sums".

Theorem 7 (Existence and uniqueness of the (p,q)-adic Haar probability measure)**.** *Let K have any quality. Then, the measure $d\mathfrak{z}$ is the unique translation-invariant (p,q)-adic measure satisfying:*

$$\int_{\mathbb{Z}_p} d\mathfrak{z} = 1 \tag{2.78}$$

As a measure, $d\mathfrak{z}$ has total variation 1, so that the triangle inequality becomes:

$$\left|\int_{\mathbb{Z}_p} f(\mathfrak{z})\, d\mathfrak{z}\right|_q \leq \|f\|_{p,q},\ \forall f \in C(\mathbb{Z}_p,K) \tag{2.79}$$

Moreover, we have the Riemann sum formula:

$$\int_{\mathbb{Z}_p} f(\mathfrak{z})\, d\mathfrak{z} \overset{K}{=} \lim_{N\to\infty} \frac{1}{p^N} \sum_{n=0}^{p^N-1} f(n),\ \forall f \in C(\mathbb{Z}_p,K) \tag{2.80}$$

as well as a formula in terms of f's vdP series:

$$\int_{\mathbb{Z}_p} f(\mathfrak{z})\, d\mathfrak{z} \overset{K}{=} \sum_{n=0}^{\infty} \frac{c_n(f)}{p^{\lambda_p(n)}},\ \forall f \in C(\mathbb{Z}_p,K) \tag{2.81}$$

Proof: Let $d\mu \in C(\mathbb{Z}_p,K)'$ be translation invariant and satisfy $\mu(\mathbb{Z}_p) = 1$. Translation invariance then forces:

$$\int_{\mathbb{Z}_p} \left[\mathfrak{z} \overset{p^n}{\equiv} k\right] d\mathfrak{z} = \mu(k + p^n\mathbb{Z}_p) = \frac{1}{p^n},\ \forall n \in \mathbb{N}_0,\ \forall k \in \mathbb{Z}/p^n\mathbb{Z} \tag{2.82}$$

This also holds in the case where K has positive characteristic, because we assumed that such an K has a residue field whose characteristic $\neq p$, which then forces $\operatorname{char}K \neq p$.

Since:

$$\left\{\left[\mathfrak{z} \overset{p^n}{\equiv} k\right] : n \in \mathbb{N}_0 : k \in \mathbb{Z}/p^n\mathbb{Z}\right\} \tag{2.83}$$

is dense in $C(\mathbb{Z}_p,K)$, this makes $C(\mathbb{Z}_p,K)$ **separable** (it contains a countable, dense subset). As a measure on a separable Banach space B is uniquely determined by what it does to a given countable dense subset of B, this proves the uniqueness of $d\mu$, which justifies our convention of denoting this $d\mu$ by $d\mathfrak{z}$.

Next, letting $f \in C(\mathbb{Z}_p, K)$, we use f's vdP series to compute the integral of $f(\mathfrak{z})\,d\mathfrak{z}$:

$$\begin{aligned}\int_{\mathbb{Z}_p} f(\mathfrak{z})\,d\mathfrak{z} &= \int_{\mathbb{Z}_p}\left(\sum_{n=0}^{\infty} c_n(f)\left[\mathfrak{z} \overset{p^{\lambda_p(n)}}{\equiv} n\right]\right)d\mathfrak{z} \\ &\overset{!}{=} \sum_{n=0}^{\infty} c_n(f)\left(\int_{\mathbb{Z}_p}\left[\mathfrak{z} \overset{p^{\lambda_p(n)}}{\equiv} n\right]d\mathfrak{z}\right) \\ &= \sum_{n=0}^{\infty}\frac{c_n(f)}{p^{\lambda_p(n)}}\end{aligned} \tag{2.84}$$

All the equalities here occur in K. The interchange of sum and integral in the step marked (!) is justified by the uniform convergence of f's vdP series in K.

Now, we show the above sum is equal to the limit of the Riemann sum given in (2.80). For this, let $N \in \mathbb{N}_0$ be arbitrary. Then:

$$\begin{aligned}\frac{1}{p^N}\sum_{n=0}^{p^N-1} f(n) &= \frac{1}{p^N}\sum_{n=0}^{p^N-1}\left(\sum_{m=0}^{\infty} c_m(f)\left[n \overset{p^{\lambda_p(m)}}{\equiv} m\right]\right) \\ (\text{swap finite sum});\ &= \sum_{m=0}^{\infty} c_m(f)\left(\frac{1}{p^N}\sum_{n=0}^{p^N-1}\left[n \overset{p^{\lambda_p(m)}}{\equiv} m\right]\right)\end{aligned}$$

Now, fix $m \in \mathbb{N}_0$, and let $N \geq \lambda_p(m)$. Then, splitting the n-sum mod $p^{\lambda_p(m)}$, we obtain:

$$\begin{aligned}\sum_{n=0}^{p^N-1}\left[n \overset{p^{\lambda_p(m)}}{\equiv} m\right] &= \sum_{n=0}^{p^{N-\lambda_p(m)}-1}\sum_{k=0}^{p^{\lambda_p(m)}-1}\left[p^{\lambda_p(m)}n + k \overset{p^{\lambda_p(m)}}{\equiv} m\right] \\ &= \sum_{n=0}^{p^{N-\lambda_p(m)}-1}\underbrace{\sum_{k=0}^{p^{\lambda_p(m)}-1}\left[k \overset{p^{\lambda_p(m)}}{\equiv} m\right]}_{1}\end{aligned}$$

Note that the inner sum evaluates to 1 because $k = m$ is the unique integer k in $\left\{0, \ldots, p^{\lambda_p(m)} - 1\right\}$ which is congruent to m mod $p^{\lambda_p(m)}$. Thus:

$$\sum_{n=0}^{p^N-1}\left[n \overset{p^{\lambda_p(m)}}{\equiv} m\right] = \sum_{n=0}^{p^{N-\lambda_p(m)}-1} 1 = p^{N-\lambda_p(m)},\ \forall N \geq \lambda_p(m) \tag{2.85}$$

Turning to the entire sum, for any $N \in \mathbb{N}_0$, we then have:

$$\begin{aligned}\frac{1}{p^N}\sum_{n=0}^{p^N-1} f(n) &= \sum_{m=0}^{\infty} c_m(f)\left(\frac{1}{p^N}\sum_{n=0}^{p^N-1}\left[n \overset{p^{\lambda_p(m)}}{\equiv} m\right]\right) \\ &= \sum_{m:\lambda_p(m)\leq N}\left(\frac{c_m(f)}{p^N}\sum_{n=0}^{p^N-1}\left[n \overset{p^{\lambda_p(m)}}{\equiv} m\right]\right) \\ &+ \sum_{m:\lambda_p(m)>N}\left(\frac{c_m(f)}{p^N}\sum_{n=0}^{p^N-1}\left[n \overset{p^{\lambda_p(m)}}{\equiv} m\right]\right)\end{aligned} \tag{2.86}$$

Since K's residue field has characteristic $\neq p$, we have that $\left|p\right|_{q}=\left|1/p\right|_{q}=1$, so the terms on the right make sense.

Now, note that summing over all m in the range $\lambda_{p}(m)>N$ forces m to have *at least* $N+1$ p-adic digits. Since $p^{(N+1)-1}=p^{N}$ is the smallest possible non-negative integer with at least $N+1$ p-adic digits, this forces:

$$\{m:\lambda_{p}(m)>N\}=\left\{m:m\geq p^{N}\right\} \tag{2.87}$$

In particular, we have that:

$$\lambda_{p}(m)>N\Rightarrow\sum_{n=0}^{p^{N}-1}\left[n\overset{p^{\lambda_{p}(m)}}{\equiv}m\right]=0 \tag{2.88}$$

which leaves us with:

$$\frac{1}{p^{N}}\sum_{n=0}^{p^{N}-1}f(n)=\sum_{m:\lambda_{p}(m)\leq N}\left(\frac{c_{m}(f)}{p^{N}}\sum_{n=0}^{p^{N}-1}\left[n\overset{p^{\lambda_{p}(m)}}{\equiv}m\right]\right) \tag{2.89}$$

And so:

$$\begin{aligned}
\lim_{N\rightarrow\infty}\frac{1}{p^{N}}\sum_{n=0}^{p^{N}-1}f(n) &\overset{K}{=}\lim_{N\rightarrow\infty}\sum_{m:\lambda_{p}(m)\leq N}\left(\frac{c_{m}(f)}{p^{N}}\sum_{n=0}^{p^{N}-1}\left[n\overset{p^{\lambda_{p}(m)}}{\equiv}m\right]\right)\\
(\text{Use }(2.85))\,; &\overset{K}{=}\lim_{N\rightarrow\infty}\sum_{m:\lambda_{p}(m)\leq N}\left(\frac{c_{m}(f)}{p^{N}}\cdot p^{N-\lambda_{p}(m)}\right)\\
&\overset{K}{=}\lim_{N\rightarrow\infty}\sum_{m:\lambda_{p}(m)\leq N}\frac{c_{m}(f)}{p^{\lambda_{p}(m)}}\\
&\overset{K}{=}\sum_{m=0}^{\infty}\frac{c_{m}(f)}{p^{\lambda_{p}(m)}}\\
(\text{Use }(2.84))\,; &\overset{K}{=}\int_{\mathbb{Z}_{p}}f(\mathfrak{z})\,d\mathfrak{z}
\end{aligned}$$

For the triangle inequality, we write:

$$\begin{aligned}
\|d\mathfrak{z}\| &= \sup_{\substack{f\in C(\mathbb{Z}_{p},K)\\ \|f\|_{p,q}\leq1}}\left|\int_{\mathbb{Z}_{p}}f(\mathfrak{z})\,d\mathfrak{z}\right|_{q}\\
&= \sup_{\substack{f\in C(\mathbb{Z}_{p},K)\\ \|f\|_{p,q}\leq1}}\left|\sum_{m=0}^{\infty}\frac{c_{m}(f)}{p^{\lambda_{p}(m)}}\right|_{q}\\
&\leq \sup_{\substack{f\in C(\mathbb{Z}_{p},K)\\ \|f\|_{p,q}\leq1}}\sup_{m\geq0}\left|c_{m}(f)\right|_{q}
\end{aligned}$$

Since f is continuous, $\left|c_{m}(f)\right|_{q}\rightarrow0$ as $m\rightarrow\infty$, and thus the ultrametric inequality gives:

$$\sup_{m\geq0}\left|c_{m}(f)\right|_{q}=\|f\|_{p,q} \tag{2.90}$$

Thus:

$$\left\|d\mathfrak{z}\right\|\leq\sup_{\substack{f\in C\left(\mathbb{Z}_{p},K\right)\\ \left\|f\right\|_{p,q}\leq1}}\sup_{m\geq0}\left|c_{m}\left(f\right)\right|_{q}=\sup_{\substack{f\in C\left(\mathbb{Z}_{p},K\right)\\ \left\|f\right\|_{p,q}\leq1}}\left\|f\right\|_{p,q}=1 \tag{2.91}$$

Since the constant function 1 is continuous, we have:

$$\left\|d\mathfrak{z}\right\|\geq\int_{\mathbb{Z}_{p}}d\mathfrak{z}=1 \tag{2.92}$$

So, $1\leq\left\|d\mathfrak{z}\right\|\leq1$, which forces $\left\|d\mathfrak{z}\right\|=1$.

Q.E.D.

Remark 8. As a result of this formula, unless specifically stated otherwise, when we say $f:\mathbb{Z}_{p}\rightarrow K$ is **integrable**, we mean integrable with respect to $d\mathfrak{z}$.

Remark 9. On the other hand, when $p=q$, *there is no Haar measure* Indeed:

$$\left|\int_{\mathbb{Z}_{p}}\left[\mathfrak{z}\overset{p^{n}}{\equiv}k\right]d\mathfrak{z}\right|_{p}=p^{n} \tag{2.93}$$

and so, in this case:

$$\left\|d\mathfrak{z}\right\|=\sup_{\substack{f\in C\left(\mathbb{Z}_{p},K\right)\\ \left\|f\right\|_{p,p}\leq1}}\left|\int_{\mathbb{Z}_{p}}f\left(\mathfrak{z}\right)d\mathfrak{z}\right|_{p}\geq\sup_{n\geq0}\left|\int_{\mathbb{Z}_{p}}\left[\mathfrak{z}\overset{p^{n}}{\equiv}0\right]d\mathfrak{z}\right|_{p}=\sup_{n\geq0}p^{n}=\infty \tag{2.94}$$

and so $d\mathfrak{z}\notin C\left(\mathbb{Z}_{p},K\right)'$ whenever K is a p-adic field. Indeed, it is a standard fact of p-adic analysis that, when K is a p-adic field, the only translation-invariant measure in $C\left(\mathbb{Z}_{p},K\right)'$ is the zero measure [15]!

Next, we have the change of variables formula for integrals:

Proposition 6 (**Change of variables formula**). *Let $\mathfrak{a},\mathfrak{b}\in\mathbb{Z}_{p}$, with $\mathfrak{a}\neq0$. Then, for all $f\in C\left(\mathbb{Z}_{p},K\right)$:*

$$\int_{\mathbb{Z}_{p}}f\left(\mathfrak{a}\mathfrak{z}+\mathfrak{b}\right)d\mathfrak{z}\overset{K}{=}\frac{1}{\left|\mathfrak{a}\right|_{p}}\int_{\mathfrak{a}\mathbb{Z}_{p}+\mathfrak{b}}f\left(\mathfrak{z}\right)d\mathfrak{z} \tag{2.95}$$

We also have a decomposition formula:

Proposition 7 (**Decomposition formula**). *Let $n\geq0$. Then, for all $f\in C\left(\mathbb{Z}_{p},K\right)$:*

$$\int_{\mathbb{Z}_{p}}f\left(\mathfrak{z}\right)d\mathfrak{z}=\sum_{k=0}^{p^{n}-1}\int_{k+p^{n}\mathbb{Z}_{p}}f\left(\mathfrak{z}\right)d\mathfrak{z}=\sum_{k=0}^{p^{n}-1}\int_{\mathbb{Z}_{p}}\left[\mathfrak{z}\overset{p^{n}}{\equiv}k\right]f\left(\mathfrak{z}\right)d\mathfrak{z} \tag{2.96}$$

Using the change of variables formula, we also have:

$$\int_{\mathbb{Z}_{p}}f\left(\mathfrak{z}\right)d\mathfrak{z}=\frac{1}{p^{n}}\sum_{k=0}^{p^{n}-1}\int_{\mathbb{Z}_{p}}f\left(p^{n}\mathfrak{z}+k\right)d\mathfrak{z} \tag{2.97}$$

Remark 10. One of the most important applications of **Proposition 7** is for integrating locally constant functions. In particular, if $f:\mathbb{Z}_p\rightarrow K$ is locally constant mod p^N, then:

$$\int_{\mathbb{Z}_p} f(\mathfrak{z})\,d\mathfrak{z}=\frac{1}{p^N}\sum_{n=0}^{p^N-1} f(n) \tag{2.98}$$

The proof of this follows from writing $f(\mathfrak{z})$ as the sum of the mod p^N indicator functions and integrating term-by-term.

It cannot be overemphasized that *all of these formulas hold independently of the quality of K.* In our work, archimedean and non-archimedean analysis sit side by side, treated with exactly the same formalism, notation, and algebra. "Standard" treatments of abstract harmonic analysis (ex: [8, 9, 14, 26]) and non-archimedean functional analysis (ex: [13, 16, 18]) cannot rise to this level of impartiality; the technical machinery the respective subjects require in order to define measures and integrable functions causes divergences between the two subjects almost from the very beginning. It is only because of the van der Put basis that we can put these two worlds of analysis on equal footing. Indeed, a careful examination of **Theorem 7**'s proof of the Haar measure's characterization and the Riemann sum formula (2.80) reveals the underlying arguments to be *purely algebraic*. The only analysis used is in the uniform convergence of the van der Put series, which is really just an algebraic property in disguise: here, it is one of the properties that $C(\mathbb{Z}_p,K)$ enjoys as a Banach space.

Differences between the archimedean and non-archimedean cases only appear when we start doing *analysis* with our formulas: considering limits, absolute values, and convergence. Case in point, the triangle inequality. When K is non-archimedean), it is an unfortunate reality of (p,q)-adic analysis that the triangle inequality (2.79) that we got in **Theorem 7** cannot be improved. Indeed, the non-archimedean analogue of the triangle inequality for integrals in real analysis is a pale imitation of its classical counterpart.

Remark 11 (**¡WARNING!**). If K is non-archimedean and $f:\mathbb{Z}_p\rightarrow K$ is integrable, the *real-valued* function $\mathfrak{z}\mapsto|f(\mathfrak{z})|_q$ is integrable with respect to the *real-valued* Haar probability measure on $\mathbb{Z}_p$. However, *there is no general relation between the quantities* $\left|\int_{\mathbb{Z}_p} f(\mathfrak{z})\,d\mathfrak{z}\right|_q$ *and* $\int_{\mathbb{Z}_p}|f(\mathfrak{z})|_q\,d\mathfrak{z}$, where the $d\mathfrak{z}$ in $\left|\int_{\mathbb{Z}_p} f(\mathfrak{z})\,d\mathfrak{z}\right|_q$ is the q-adic-valued Haar probability measure on $\mathbb{Z}_p$, and the $d\mathfrak{z}$ in $\int_{\mathbb{Z}_p}|f(\mathfrak{z})|_q\,d\mathfrak{z}$ is the *real-valued* Haar probability measure on $\mathbb{Z}_p$. Indeed, there exist $f\in C(\mathbb{Z}_p,\mathbb{Q}_q)$ so that:

$$\left|\int_{\mathbb{Z}_p} f(\mathfrak{z})\,d\mathfrak{z}\right|_q>\int_{\mathbb{Z}_p}|f(\mathfrak{z})|_q\,d\mathfrak{z} \tag{2.99}$$

For a concrete example, let $a\in\mathbb{Q}$. Then:

$$\left|\int_{\mathbb{Z}_p}\left(a\left[\mathfrak{z}\overset{p}{\equiv}0\right]\right)d\mathfrak{z}\right|_q=\left|\frac{a}{p}\right|_q \tag{2.100}$$

On the other hand, note that:

$$\left|a\left[\mathfrak{z}\overset{p}{\equiv}0\right]\right|_q=|a|_q\left[\mathfrak{z}\overset{p}{\equiv}0\right],\ \forall\mathfrak{z}\in\mathbb{Z}_p \tag{2.101}$$

Hence:

$$\int_{\mathbb{Z}_p}\left|a\left[\mathfrak{z}\overset{p}{\equiv}0\right]\right|_q d\mathfrak{z}=\int_{\mathbb{Z}_p}|a|_q\left[\mathfrak{z}\overset{p}{\equiv}0\right]d\mathfrak{z}=\frac{|a|_q}{p} \tag{2.102}$$

Choosing $a=1$, we have that $\left|a/p\right|_{q}=1$ (because $p\neq q$) and $\left|a\right|_{q}/p=1/p$, and so:

$$\left|\int_{\mathbb{Z}_{p}}\left[\mathfrak{z}\overset{p}{\equiv}0\right]d\mathfrak{z}\right|_{q}=1>\frac{1}{p}=\int_{\mathbb{Z}_{p}}\left|\left[\mathfrak{z}\overset{p}{\equiv}0\right]\right|_{q}d\mathfrak{z} \tag{2.103}$$

Note 3. Combining **Theorem 6** with **Theorem 7** then proves the **Fundamental Theorem of** (p,q)**-adic Analysis (Theorem 4**).

Definition 9. With our notations, the formalism for defining and computing the (p,q)-adic Fourier transform is completely identical to the Fourier transform of complex-valued functions on $\mathbb{Z}_{p}$ as done in [9, 14]. Since our characters continuous (p,q)-adic functions of $\mathfrak{z}$, given any $t\in\hat{\mathbb{Z}}_{p}$, we then define the measure $e^{-2\pi i\{t\mathfrak{z}\}_{p}}d\mathfrak{z}$ as the K-valued functional which, when fed $f\in C\left(\mathbb{Z}_{p},K\right)$, integrates the continuous function $f\left(\mathfrak{z}\right)e^{-2\pi i\{t\mathfrak{z}\}_{p}}$ with respect to $d\mathfrak{z}$. With this, we define the (p,q)**-adic Fourier transform** on $\mathbb{Z}_{p}$ by the formula:

$$\mathscr{F}\left\{f\right\}\left(t\right)\overset{\textrm{def}}{=}\int_{\mathbb{Z}_{p}}f\left(\mathfrak{z}\right)e^{-2\pi i\{t\mathfrak{z}\}_{p}}d\mathfrak{z},\ \forall t\in\hat{\mathbb{Z}}_{p} \tag{2.104}$$

Writing $\hat{f}\left(t\right)$ to denote $\mathscr{F}\left\{f\right\}\left(t\right)$, we have that $\hat{f}$ is a function $\hat{\mathbb{Z}}_{p}\rightarrow K$. The **Fourier series** of f is then:

$$\sum_{t\in\hat{\mathbb{Z}}_{p}}\hat{f}\left(t\right)e^{2\pi i\{t\mathfrak{z}\}_{p}} \tag{2.105}$$

By **Theorem 7**, we have that:

$$\hat{f}\left(t\right)\overset{K}{=}\lim_{N\rightarrow\infty}\frac{1}{p^{N}}\sum_{n=0}^{p^{N}-1}f\left(n\right)e^{-2\pi int},\ \forall t\in\hat{\mathbb{Z}}_{p} \tag{2.106}$$

Remark 12. The main Fourier integral worth knowing is:

$$\int_{\mathbb{Z}_{p}}e^{-2\pi i\{t\mathfrak{z}\}_{p}}d\mathfrak{z}\overset{K}{=}\mathbf{1}_{0}\left(t\right),\ \forall t\in\hat{\mathbb{Z}}_{p} \tag{2.107}$$

that is to say, the Fourier transform of the constant function 1 is $\mathbf{1}_{0}\left(t\right)$. With this, the Fourier transforms of $\left[\mathfrak{z}\overset{p^{n}}{\equiv}k\right]$ may be computed using **Proposition 4**. Fourier integrals can also be computed directly using the Riemann sum formula (2.80).

The **Riemann-Lebesgue Lemma** is also universal for q-adic fields, regardless of quality, provided that we state it for *continuous* functions, rather than L^{1} functions.

Lemma 2 (The Riemann-Lebesgue Lemma)**.** *Regardless of the quality of* K*, if* $f\in C\left(\mathbb{Z}_{p},K\right)$*, then:*

$$\lim_{\left|t\right|_{p}\rightarrow\infty}\hat{f}\left(t\right)\overset{K}{=}0 \tag{2.108}$$

Proof: If K is archimedean the proof is classical (see [8]). If K is non-archimedean, the decay follows from **Corollary 6**.

Q.E.D.

Using this and the fundamental theorem, we get:

Corollary 1. *The* (p,q)*-adic Fourier transform* $\mathscr{F}:C\left(\mathbb{Z}_{p},K\right)\rightarrow c_{0}\left(\hat{\mathbb{Z}}_{p},K\right)$ *is an isometric isomorphism of Banach spaces.*

Proof: **Theorem 6** shows $\mathscr{F}$ is an isomorphism of Banach spaces. That $\mathscr{F}$ is an isometry follows from letting $f \in C(\mathbb{Z}_p, K)$ and applying the ultrametric inequality to the norm:

$$\|f\|_{p,q} = \sup_{\mathfrak{z} \in \mathbb{Z}_p} |f(\mathfrak{z})|_q = \sup_{\mathfrak{z} \in \mathbb{Z}_p} \left| \sum_{t \in \hat{\mathbb{Z}}_p} \hat{f}(t) e^{2\pi i \{t\mathfrak{z}\}_p} \right|_q \leq \sup_{\mathfrak{z} \in \mathbb{Z}_p} \sup_{t \in \hat{\mathbb{Z}}_p} \left| \hat{f}(t) \right|_q = \left\| \hat{f} \right\|_{p,q} \tag{2.109}$$

and:

$$\left\| \hat{f} \right\|_{p,q} = \sup_{t \in \hat{\mathbb{Z}}_p} \left| \hat{f}(t) \right|_q = \sup_{t \in \hat{\mathbb{Z}}_p} \left| \int_{\mathbb{Z}_p} f(\mathfrak{z}) e^{-2\pi i \{t\mathfrak{z}\}_p} d\mathfrak{z} \right|_q \leq \sup_{t \in \hat{\mathbb{Z}}_p} \sup_{\mathfrak{z} \in \mathbb{Z}_p} |f(\mathfrak{z})|_q = \|f\|_{p,q} \tag{2.110}$$

So, $\|f\|_{p,q}$ and $\left\| \hat{f} \right\|_{p,q}$ are less than or equal to one another, and hence, equal.

Q.E.D.

When K is non-archimedean, since every $f \in C(\mathbb{Z}_p, K)$ can be uniquely written as a uniformly convergent Fourier series:

$$f(\mathfrak{z}) \overset{K}{=} \sum_{t \in \hat{\mathbb{Z}}_p} \hat{f}(t) e^{2\pi i \{t\mathfrak{z}\}_p} \tag{2.111}$$

we see that the characters $\left\{ \mathfrak{z} \mapsto e^{2\pi i \{t\mathfrak{z}\}_p} \right\}_{t \in \hat{\mathbb{Z}}_p}$ form a basis for $C(\mathbb{Z}_p, K)$. Consequently, purely by linear algebra, any measure $d\mu \in C(\mathbb{Z}_p, K)'$ is completely determined by what it does to the continuous functions $\left\{ \mathfrak{z} \mapsto e^{2\pi i \{t\mathfrak{z}\}_p} \right\}_{t \in \hat{\mathbb{Z}}_p}$. This gives us the **Fourier-*Stieltjes* transform**.

Definition 10. The (p,q)**-adic Fourier-Stieltjes transform** of a measure $d\mu \in C(\mathbb{Z}_p, K)'$ is defined by:

$$\mathscr{F}\{d\mu\}(t) \overset{\text{def}}{=} \int_{\mathbb{Z}_p} e^{-2\pi i \{t\mathfrak{z}\}_p} d\mu(\mathfrak{z}), \ \forall t \in \hat{\mathbb{Z}}_p \tag{2.112}$$

that is, as the image of the continuous function $\mathfrak{z} \mapsto e^{-2\pi i \{t\mathfrak{z}\}_p}$ under the measure $d\mu$. We write $\hat{\mu}(t)$ to denote $\mathscr{F}\{d\mu\}(t)$; this is a function $\hat{\mathbb{Z}}_p \to K$.

Since the characters form a basis for $C(\mathbb{Z}_p, K)$ in the non-archimedean case, given any $f \in C(\mathbb{Z}_p, K)$, we can compute f's image under a measure $d\mu$ through linear algebra. This gives the **Parseval-Plancherel identity**:

Theorem 8 (**Parseval-Plancherel Identity**). *Let $d\mu \in C(\mathbb{Z}_p, K)'$. Then:*

$$\int_{\mathbb{Z}_p} f(\mathfrak{z}) d\mu(\mathfrak{z}) \overset{K}{=} \sum_{t \in \hat{\mathbb{Z}}_p} \hat{f}(t) \hat{\mu}(-t), \ \forall f \in C(\mathbb{Z}_p, K) \tag{2.113}$$

Moreover, for any $f, g \in C(\mathbb{Z}_p, K)$:

$$\int_{\mathbb{Z}_p} f(\mathfrak{z}) g(\mathfrak{z}) d\mathfrak{z} \overset{K}{=} \sum_{t \in \hat{\mathbb{Z}}_p} \hat{f}(t) \hat{g}(-t) \tag{2.114}$$

Proof:

$$\begin{aligned}\int_{\mathbb{Z}_p} f(\mathfrak{z})\,d\mu(\mathfrak{z}) &\overset{K}{=} \int_{\mathbb{Z}_p}\sum_{t\in\hat{\mathbb{Z}}_p}\hat{f}(t)\,e^{2\pi i\{t\mathfrak{z}\}_p}d\mu(\mathfrak{z})\\ &\overset{K}{=}\sum_{t\in\hat{\mathbb{Z}}_p}\hat{f}(t)\left(\int_{\mathbb{Z}_p}e^{2\pi i\{t\mathfrak{z}\}_p}d\mu(\mathfrak{z})\right)\\ &=\sum_{t\in\hat{\mathbb{Z}}_p}\hat{f}(t)\,\hat{\mu}(-t)\end{aligned}$$

Note that the interchange of sum and integral in the second line is justified by the uniform convergence of f's Fourier series. We get the case with g by considering the measure $d\mu(\mathfrak{z}) = g(\mathfrak{z})\,d\mathfrak{z}$.

Q.E.D.

From the above, we see that given a measure $d\mu$ and a continuous (p,q)-adic function f, the recipe for the image of f under $d\mu$ is:

$$\begin{aligned}C(\mathbb{Z}_p,K)&\to K\\ f&\mapsto\sum_{t\in\hat{\mathbb{Z}}_p}\hat{f}(t)\,\hat{\mu}(-t)\end{aligned}\tag{2.115}$$

Conversely, as a consequence of the **Fundamental Theorem**, it immediately follows that *for any* function $\hat{\mu}\in B\left(\hat{\mathbb{Z}}_p,K\right)$, the equation (2.115) defines an element of $C(\mathbb{Z}_p,K)'$. From this, we obtain:

Theorem 9. *If K is non-archimedean, the (p,q)-adic Fourier-Stieltjes transform $\mathscr{F}: C(\mathbb{Z}_p,K)'\to B\left(\hat{\mathbb{Z}}_p,K\right)$ is an isometric isomorphism of Banach spaces.*

We can upgrade these result to an isometric isomorphisms of Banach *algebras* by introducing the operation of convolution.

Definition 11. We use $*$ to define the **convolution operation** for (p,q)-adic functions on $\mathbb{Z}_p$ and $\hat{\mathbb{Z}}_p$. These are given by:

$$(f*g)(\mathfrak{z})\overset{\text{def}}{=}\int_{\mathbb{Z}_p}f(\mathfrak{z}-\mathfrak{y})\,g(\mathfrak{y})\,d\mathfrak{y}\tag{2.116}$$

$$\left(\hat{f}*\hat{g}\right)(t)\overset{\text{def}}{=}\sum_{s\in\hat{\mathbb{Z}}_p}\hat{f}(t-s)\,\hat{g}(s)\tag{2.117}$$

Proposition 8. *$*$ is a commutative K-bilinear operation, just like in the classical case, and we have:*

$$\begin{gathered}\alpha(f*g)(\mathfrak{z})=(\alpha f*g)(\mathfrak{z})=(f*\alpha g)(\mathfrak{z})\\ ((f+h)*g)(\mathfrak{z})=(f*g)(\mathfrak{z})+(h*g)(\mathfrak{z})\\ (f*(g+h))(\mathfrak{z})=(f*g)(\mathfrak{z})+(f*h)(\mathfrak{z})\\ (f*g)(\mathfrak{z})=(g*f)(\mathfrak{z})\end{gathered}$$

for all $f,g,h\in C(\mathbb{Z}_p,K)$, all $\alpha\in K$, and all $\mathfrak{z}\in\mathbb{Z}_p$. The analogous results for hatted functions and $t\in\hat{\mathbb{Z}}_p$ instead of $\mathfrak{z}\in\mathbb{Z}_p$ also hold.

We can also convolve measures, both with functions, and with one another:

Definition 12. Let $d\mu \in C(\mathbb{Z}_p, K)'$, and let $f \in C(\mathbb{Z}_p, K)$. Then, we define the convolution $f * d\mu$ by:

$$(f * d\mu)(\mathfrak{z}) \overset{\text{def}}{=} \int_{\mathbb{Z}_p} f(\mathfrak{z} - \mathfrak{y})\, d\mu(\mathfrak{y}) \tag{2.118}$$

for all $\mathfrak{z} \in \mathbb{Z}_p$ for which the integral exists. If we are given another measure $d\nu \in C(\mathbb{Z}_p, K)'$, we define $d\mu * d\nu$ to be the measure given by:

$$\int_{\mathbb{Z}_p} g(\mathfrak{z})\,(d\mu * d\nu)(\mathfrak{z}) \overset{\text{def}}{=} \sum_{t \in \hat{\mathbb{Z}}_p} \hat{g}(t)\, \hat{\mu}(-t)\, \hat{\nu}(-t)\,,\ \forall g \in C(\mathbb{Z}_p, K) \tag{2.119}$$

Note that this definition does, in fact, give us a measure, by **Theorem 9**, seeing as $t \mapsto \hat{\mu}(t)\, \hat{\nu}(t)$ is in $B\left(\hat{\mathbb{Z}}_p, K\right)$.

As one would expect, the convolution of a function and a measure is again a function:

Proposition 9. *Let $d\mu \in C(\mathbb{Z}_p, K)'$, and let $f \in C(\mathbb{Z}_p, K)$. Then, $f * d\mu \in C(\mathbb{Z}_p, K)$, with:*

$$(f * d\mu)(\mathfrak{z}) = \sum_{t \in \hat{\mathbb{Z}}_p} \hat{f}(t)\, \hat{\mu}(t)\, e^{2\pi i \{t\mathfrak{z}\}_p} \tag{2.120}$$

More generally:

$$(f * g)(\mathfrak{z}) = \sum_{t \in \hat{\mathbb{Z}}_p} \hat{f}(t)\, \hat{g}(t)\, e^{2\pi i \{t\mathfrak{z}\}_p},\ \forall f, g \in C(\mathbb{Z}_p, K)$$

Proof: Since f is continuous, we can write:

$$\begin{aligned}
\int_{\mathbb{Z}_p} f(\mathfrak{z} - \mathfrak{y})\, d\mu(\mathfrak{y}) &= \int_{\mathbb{Z}_p} \left(\sum_{t \in \hat{\mathbb{Z}}_p} \hat{f}(t)\, e^{2\pi i \{t(\mathfrak{z} - \mathfrak{y})\}_p} \right) d\mu(\mathfrak{y}) \\
&\overset{!}{=} \sum_{t \in \hat{\mathbb{Z}}_p} \hat{f}(t) \left(\int_{\mathbb{Z}_p} e^{-2\pi i \{t\mathfrak{y}\}_p} d\mu(\mathfrak{y}) \right) e^{2\pi i \{t\mathfrak{z}\}_p} \\
&= \sum_{t \in \hat{\mathbb{Z}}_p} \hat{f}(t)\, \hat{\mu}(t)\, e^{2\pi i \{t\mathfrak{z}\}_p}
\end{aligned}$$

where the interchange of sum and integral in the step marked (!) is justified by the uniform convergence of the Fourier series of f, itself guaranteed by the non-archimedean quality of K and the continuity of f. Since $d\mu$ is a measure, $\hat{\mu}$ is q-adically bounded, and thus the map $t \mapsto \hat{f}(t)\, \hat{\mu}(t)$ is in $c_0\left(\hat{\mathbb{Z}}_p, K\right)$, which—by the **Fundamental Theorem**—guarantees that the Fourier series on the bottom line defines a continuous (p, q)-adic function.

The second formula follows from the first by setting $d\mu(\mathfrak{z}) = g(\mathfrak{z})\, d\mathfrak{z}$.

Q.E.D.

Like its classical counterparts, the (p, q)-adic Fourier transform satisfies the **Convolution Theorem**:

Theorem 10 (Convolution Theorem)**.** *Let K be non-archimedean, and let $f, g \in C(\mathbb{Z}_p, K)$. Then:*

$$\mathscr{F}\{f * g\}(t) = \mathscr{F}\{f\}(t) \times \mathscr{F}\{g\}(t) \tag{2.121}$$

$$\mathscr{F}\{f \times g\}(t) = (\mathscr{F}\{f\} * \mathscr{F}\{g\})(t) \tag{2.122}$$

for all $t \in \hat{\mathbb{Z}}_p$. *Additionally, for measures* $d\mu, d\nu \in C(\mathbb{Z}_p, K)'$, *the Fourier-Stieltjes transform satisfies: we have:*

$$\mathscr{F}\{d\mu * d\nu\}(t) = \mathscr{F}\{d\mu\}(t) \times \mathscr{F}\{d\nu\}(t) \tag{2.123}$$

for all $t \in \hat{\mathbb{Z}}_p$.

We then get the analogue of **Young's Inequality for Convolutions**:

Corollary 2. *Let* K *be non-archimedean. Then:*

$$\|F * G\| \leq \|F\| \, \|G\| \tag{2.124}$$

$$\left\|\hat{f} * \hat{g}\right\| \leq \left\|\hat{f}\right\| \|\hat{g}\| \tag{2.125}$$

for all $F, G \in C(\mathbb{Z}_p, K)$ *and all* $\hat{f}, \hat{g} \in c_0\left(\hat{\mathbb{Z}}_p, K\right)$. *We also have:*

$$\left|\int_{\mathbb{Z}_p} h(\mathfrak{z})\,(d\mu * d\nu)(\mathfrak{z})\right|_q \leq \left\|\hat{h}\right\| \|\hat{\mu}\| \, \|\hat{\nu}\| = \left\|\hat{h}\right\| \|\mu\| \, \|\nu\| \tag{2.126}$$

for all $h \in C(\mathbb{Z}_p, K)$ *and all* $d\mu, d\nu \in C(\mathbb{Z}_p, K)'$.

Proof: Apply norms to the identities of **Theorem 10**, then use the **Fundamental Theorem** to change $\left\|\hat{F}\right\|$ to $\|F\|$ and $\|f\|$ to $\left\|\hat{f}\right\|$, and similarly for G and $\hat{g}$. Likewise, the inequality for measures follows from **Theorem 10** and the isometry property of the Fourier-Stieltjes transform (**Theorem 9**).

Remark 13. Let K be algebraically closed and non-archimedean. With convolution and point-wise multiplication, we have two different ways of making function spaces into Banach algebras. $C(\mathbb{Z}_p, K)$ becomes a Banach algebra under either point-wise multiplication or convolution, as does. $c_0\left(\hat{\mathbb{Z}}_p, K\right)$. We denote these algebras by $(C(\mathbb{Z}_p, K), \times)$, $(C(\mathbb{Z}_p, K), *)$, and $\left(c_0\left(\hat{\mathbb{Z}}_p, K\right), \times\right)$ and $\left(c_0\left(\hat{\mathbb{Z}}_p, K\right), *\right)$, respectively. Like with spaces of real valued measures, $C(\mathbb{Z}_p, K)'$ is a Banach algebra under convolution (denoted $\left(C(\mathbb{Z}_p, K)', *\right)$), but *not* under point-wise multiplication. Similarly, $B\left(\hat{\mathbb{Z}}_p, K\right)$ is a Banach algebra under point-wise multiplication (denoted $\left(B\left(\hat{\mathbb{Z}}_p, K\right), \times\right)$), but *not* under convolution.

In summary, we have:

Theorem 11. *Let* K *be non-archimedean. Then:*

I. The Fourier transform $\mathscr{F} : (C(\mathbb{Z}_p, K), \times) \to \left(c_0\left(\hat{\mathbb{Z}}_p, K\right), *\right)$ *is an isometric isomorphism of* <u>*unital*</u> K*-Banach algebras.*

II. The Fourier transform $\mathscr{F} : (C(\mathbb{Z}_p, K), *) \to \left(c_0\left(\hat{\mathbb{Z}}_p, K\right), \times\right)$ *is an isometric isomorphism of* <u>*non-unital*</u> K*-Banach algebras.*

III. The Fourier-Stieltjes transform $\mathscr{F} : \left(C(\mathbb{Z}_p, K)', *\right) \to \left(B\left(\hat{\mathbb{Z}}_p, K\right), \times\right)$ *is an isometric isomorphism of* <u>*unital*</u> K*-Banach algebras.*

2.5. **The p-Adic Dirichlet Kernel and Radon-Nikodym Differentiation.** Because the "standard" notion of Radon-Nikodym differentiation in non-archimedean functional analysis is a bit of mouthful, we will first present our simpler, more focused version of the concept before comparing it to the literature.

We begin with the **p-adic Dirichlet kernel**.

Definition 13. Let K be any field. The **p-adic Dirichlet kernel** $D_N : \mathbb{Z}_p \to K$ is the locally constant function:

$$D_N(\mathfrak{z}) \overset{\text{def}}{=} p^N \left[\mathfrak{z} \overset{p^N}{\equiv} 0\right] \tag{2.127}$$

Remark 14. Note that K here can be any field, either archimedean or non-archimedean. However, it is only when K is non-archimedean that the Fourier theory gives D_N the exceedingly nice properties described below.

Assumption 4. *THROUGHOUT THE REST OF THIS SUBSECTION, UNLESS STATED OTHERWISE, WE ASSUME K IS NON-ARCHIMEDEAN.*

To explain the name, we first need to prove D_N's most important property.

Proposition 10. *Let $f \in C(\mathbb{Z}_p, K)$ and let $d\mu \in C(\mathbb{Z}_p, K)'$. Then, for all $N \geq 0$:*

$$(D_N * f)(\mathfrak{z}) = \sum_{|t|_p \leq p^N} \hat{f}(t)\, e^{2\pi i \{t\mathfrak{z}\}_p},\ \forall \mathfrak{z} \in \mathbb{Z}_p \tag{2.128}$$

$$(D_N * d\mu)(\mathfrak{z}) = \sum_{|t|_p \leq p^N} \hat{\mu}(t)\, e^{2\pi i \{t\mathfrak{z}\}_p},\ \forall \mathfrak{z} \in \mathbb{Z}_p \tag{2.129}$$

*In particular $\lim_{N\to\infty} (D_N * f)(\mathfrak{z})$ converges in K to $f(\mathfrak{z})$ uniformly with respect to $\mathfrak{z} \in \mathbb{Z}_p$; that is to say, D_N is an **approximate identity** of $C(\mathbb{Z}_p, K)$.*

Proof: Writing:

$$p^N \left[\mathfrak{z} \overset{p^N}{\equiv} 0\right] = \sum_{|t|_p \leq p^N} e^{2\pi i \{t\mathfrak{z}\}_p}$$

we see that the Fourier transform of D_N is $\mathbf{1}_0\left(p^N t\right)$, the indicator function of the set of all $t \in \hat{\mathbb{Z}}_p$ with $|t|_p \leq p^N$. Thus, by **Proposition 9**:

$$(D_N * f)(\mathfrak{z}) = \sum_{t \in \hat{\mathbb{Z}}_p} \mathbf{1}_0(p^n t)\, \hat{f}(t)\, e^{2\pi i \{t\mathfrak{z}\}_p} = \sum_{|t|_p \leq p^N} \hat{f}(t)\, e^{2\pi i \{t\mathfrak{z}\}_p}$$

for all $f \in C(\mathbb{Z}_p, K)$.

The formula for $D_N * d\mu$ follows similarly.

The convergence in K of $\lim_{N\to\infty} (D_N * f)(\mathfrak{z})$ to $f(\mathfrak{z})$ uniformly with respect to $\mathfrak{z} \in \mathbb{Z}_p$ then follows from the **Fundamental Theorem**.

Q.E.D.

Remark 15. As for the name we have given D_N, recall from classical Fourier Analysis (viz. [25]) that, for $N \in \mathbb{N}_0$, the N**th Dirichlet kernel** is defined by the function:

$$D_N(t) \overset{\text{def}}{=} \sum_{n=-N}^{N} e^{2\pi i n x} = \frac{\sin((2N+1)\pi t)}{\sin(\pi t)} \tag{2.130}$$

Convolving a function $f:\mathbb{R}/\mathbb{Z}\rightarrow\mathbb{C}$ with D_N yields the Nth symmetric partial sum of f's Fourier series:

$$(D_N * f)(t) = \int_0^1 D_N(t-x) f(x)\,dx = \sum_{n=-N}^{N} \hat{f}(n) e^{2\pi i n t} \tag{2.131}$$

where, of course:

$$\hat{f}(n) = \int_0^1 f(t) e^{-2\pi i n t} dt \tag{2.132}$$

Unlike its p-adic counterpart, the classical Dirichlet kernel is *not* an approximate identity on the space of continuous functions $\mathbb{R}/\mathbb{Z}\rightarrow\mathbb{C}$ [25]. This is due to the well-known estimate:

$$\int_0^1 |D_N(t)|\,dt \geq \frac{\ln(2N+1)}{\pi^2},\ \forall N\in\mathbb{N}_0 \tag{2.133}$$

Indeed, one of the conditions a family of functions $\{K_N\}_{N\geq 0}$ on $\mathbb{R}/\mathbb{Z}$ must satisfy in order for $f * K_N$ to converge uniformly to f for any continuous $f:\mathbb{R}/\mathbb{Z}\rightarrow\mathbb{C}$ must satisfy the normalization condition:

$$\int_0^1 |K_N(t)|\,dt = 1,\ \forall N\in\mathbb{N}_0 \tag{2.134}$$

With the p-adic Dirichlet kernel in hand, we can proceed directly to Radon-Nikodym differentiation. Classically, the Radon-Nikodym derivative of a measure arises in the circle of ideas and constructions surrounding the **Lebesgue Differentiation Theorem** and the **Hardy-Littlewood maximal operator**. For simplicity, let us work on $\mathbb{R}^d$ for $d\in\mathbb{N}_1$. Following Folland ([7]), upon letting $L^1_{\text{loc}}\left(\mathbb{R}^d,\mathbb{C}\right)$ denote the space of locally integrable functions $\mathbb{R}^d\rightarrow\mathbb{C}$, for every $\mathbf{x}\in\mathbb{R}^d$ and every radius $r>0$, we define:

$$A_r\{f\}(\mathbf{x}) \overset{\text{def}}{=} \frac{1}{|B(\mathbf{x},r)|}\int_{B(\mathbf{x},r)} f(\mathbf{y})\,d\mathbf{y},\ \forall f\in L^1_{\text{loc}}\left(\mathbb{R}^d,\mathbb{C}\right) \tag{2.135}$$

where $B(\mathbf{x},r)$ is the open ball in $\mathbb{R}^d$ centered at $\mathbf{x}$ of radius r, and where $|B(\mathbf{x},r)|$ is the d-dimensional volume of $B(\mathbf{x},r)$. The integral here is taken over B with respect to the Lebesgue measure on $\mathbb{R}^d$. A slight modification of this formula yields the famous **Hardy-Littlewood maximal operator**:

$$H\{f\}(\mathbf{x}) \overset{\text{def}}{=} \sup_{r>0} A_r\{|f|\}(\mathbf{x}) \tag{2.136}$$

With a little help from the **Vitali Covering Lemma**, one can establish the ever-useful **Hardy Littlewood Maximal Inequality**, which can then, in turn, be used to prove the **Lebesgue Differentiation Theorem**:

Theorem 12 (Lebesgue Differentiation Theorem [7])**.** *Let* $f\in L^1_{loc}\left(\mathbb{R}^d,\mathbb{C}\right)$. *Then:*

$$\lim_{r\rightarrow 0} A_r\{f\}(\mathbf{x}) \overset{\mathbb{C}}{=} f(\mathbf{x}) \tag{2.137}$$

for almost every $\mathbf{x}\in\mathbb{R}^d$.

Given a complex measure $d\mu$ on $\mathbb{R}^d$ which is **absolutely continuous** with respect to the Lebesgue measure (recall, this means every Lebesgue-measurable subset of $\mathbb{R}^d$ with zero Lebesgue measure also has zero μ measure), if we define:

$$A_r\left\{d\mu\right\}(\mathbf{x}) \overset{\textrm{def}}{=} \frac{1}{\left|B(\mathbf{x},r)\right|}\int_{B(\mathbf{x},r)} d\mu(\mathbf{y}) \tag{2.138}$$

the limit of $A_r\left\{d\mu\right\}(\mathbf{x})$ as $r \to \infty$ will exist for almost every $\mathbf{x}$. The function defined by this almost-everywhere limit is then precisely the Radon-Nikodym derivative of $d\mu$. Of course, the Lebesgue Differentiation Theorem holds on many measure spaces, not just Euclidean ones. In particular, there is a version for functions $\mathbb{Z}_p \to \mathbb{C}$. Amusingly, the correct formula for the p-adic analogue of formula (2.137) can be deduced just by re-writing $A_r\left\{f\right\}$ in p-adic terms.

In $\mathbb{Z}_p$, the analogue of a ball of radius r centered at $\mathbf{x}$ is the set $\mathfrak{z} + p^N\mathbb{Z}_p$, for some $\mathfrak{z} \in \mathbb{Z}_p$ and $N \geq 0$. The analogue of the Lebesgue measure over $\mathbb{Z}_p$ is our Haar measure, which assigns measure $1/p^N$ to $\mathfrak{z} + p^N\mathbb{Z}_p$. Thus, for $f : \mathbb{Z}_p \to \mathbb{C}$ (a (p,∞)**-adic function**), our candidate for the p-adic analogue of the averaging operator (2.135) is:

$$A_N\left\{f\right\}(\mathfrak{z}) \overset{\textrm{def}}{=} \frac{1}{1/p^N}\int_{\mathfrak{z}+p^N\mathbb{Z}_p} f(\mathfrak{y})\,d\mathfrak{y} \tag{2.139}$$

where $d\mathfrak{y}$ is the real-valued p-adic Haar probability measure. Using our knowledge of p-adic integrals, we immediately note that:

$$\int_{\mathfrak{z}+p^N\mathbb{Z}_p} f(\mathfrak{y})\,d\mathfrak{y} = \int_{\mathbb{Z}_p}\left[\mathfrak{y} \overset{p^N}{\equiv} \mathfrak{z}\right] f(\mathfrak{y})\,d\mathfrak{y} \tag{2.140}$$

and hence:

$$A_N\left\{f\right\}(\mathfrak{z}) = p^N\int_{\mathbb{Z}_p}\left[\mathfrak{y} - \mathfrak{z} \overset{p^N}{\equiv} 0\right] f(\mathfrak{y})\,d\mathfrak{y} = \left(D_N * f\right)(\mathfrak{z}) \tag{2.141}$$

In other words **Proposition 10**'s assertion that D_N is an approximate identity in (p,q)-adic analysis is precisely the (p,q)-adic incarnation of the Lebesgue Differentiation Theorem! Unsurprisingly, there is also a (p,∞)-adic analogue; see Sections 2.8 - 2.9 of [6], where the result is proven for finite Borel measures on separable, locally compact ultrametric spaces such as $\mathbb{Z}_p$. In the results given below, $L^1(\mathbb{Z}_p,\mathbb{C})$ denotes the space of complex-valued functions on $\mathbb{Z}_p$ whose absolute values are integrable with respect to $\mathbb{Z}_p$'s real-valued Haar probability measure.

Theorem 13 ((p,∞)-adic Lebesgue Differentiation Theorem). *In particular, given any $f \in L^1(\mathbb{Z}_p,\mathbb{C})$:*

$$\lim_{N\to\infty}\left(D_N * f\right)(\mathfrak{z}) \overset{\mathbb{C}}{=} \lim_{N\to\infty} p^N\int_{\mathbb{Z}_p}\left[\mathfrak{y} \overset{p^N}{\equiv} \mathfrak{z}\right] f(\mathfrak{y})\,d\mathfrak{y} = f(\mathfrak{z}) \tag{2.142}$$

for almost every $\mathfrak{z} \in \mathbb{Z}_p$; in particular, the limit holds for all $\mathfrak{z}$ at which f is continuous.

Because of this, we have:

Corollary 3. *Let $d\mu$ be a finite, complex-valued Borel measure on $\mathbb{Z}_p$ which is absolutely continuous with respect to $\mathbb{Z}_p$'s real-valued Haar probability measure. Then, there is an $f \in L^1(\mathbb{Z}_p,\mathbb{C})$ so that $d\mu(\mathfrak{z}) = f(\mathfrak{z})\,d\mathfrak{z}$, where $d\mathfrak{z}$ is $\mathbb{Z}_p$'s real-valued Haar probability measure. Moreover:*

$$\lim_{N\to\infty}\left(D_N * d\mu\right)(\mathfrak{z}) \overset{\mathbb{C}}{=} f(\mathfrak{z})$$

for almost every $\mathfrak{z} \in \mathbb{Z}_p$; in particular, the limit holds for all $\mathfrak{z}$ at which f is continuous. We then call f the ***Radon-Nikodym derivative*** *of $d\mu$ with respect to $\mathbb{Z}_p$'s real-valued Haar probability measure.*

This motivates the following definition:

Definition 14. Let $d\nu \in C(\mathbb{Z}_p, K)'$. We say $d\nu$ is **(Radon-Nikodym) differentiable with respect to the Haar measure** at $\mathfrak{z} \in \mathbb{Z}_p$ whenever the limit:

$$\lim_{N\to\infty} (D_N * d\nu)(\mathfrak{z}) \tag{2.143}$$

exists in K, in which case, we call the limit the **(Radon-Nikodym) derivative of** $d\nu$ at $\mathfrak{z}$ **with respect to the Haar measure**. We say $\mathfrak{z}$ is a **regular point of** $d\nu$ whenever $d\nu$ is Radon-Nikodym differentiable at $\mathfrak{z}$.

More generally, given $d\nu \in C(\mathbb{Z}_p, K)'$, we say $d\nu$ is (**Radon-Nikodym) differentiable with respect to** $d\mu$ at $\mathfrak{z} \in \mathbb{Z}_p$ whenever the limit:

$$\lim_{N\to\infty} \frac{\int_{\mathfrak{z}+p^N\mathbb{Z}_p} d\nu(\mathfrak{x})}{\int_{\mathfrak{z}+p^N\mathbb{Z}_p} d\mu(\mathfrak{y})} \tag{2.144}$$

exists in K, in which case, we call the value of the limit the (**Radon-Nikodym) derivative of** $d\nu$ **with respect to** $d\mu$ We say $\mathfrak{z}$ is a μ**-regular point of** $d\nu$ whenever $d\nu$ is Radon-Nikodym differentiable with respect to $d\mu$ at $\mathfrak{z}$.

Assumption 5. *From here on out, unless stated otherwise, all derivatives and are Radon-Nikodym derivatives with respect to the (p,q)-adic Haar probability measure, and likewise for regular points.*

With all that said and done, we can finally address the similarities and differences between our treatment of Radon-Nikodym differentiation and the treatments common in the literature. It isn't too difficult to see that our limiting procedure:

$$\lim_{N\to\infty} \frac{\int_{\mathfrak{z}+p^N\mathbb{Z}_p} d\nu(\mathfrak{x})}{\int_{\mathfrak{z}+p^N\mathbb{Z}_p} d\mu(\mathfrak{y})} \tag{2.145}$$

is a specific case of the more general construction given by Schikhof in **Theorem 2.2** of [17]. Insofar as it concerns measures on $\mathbb{Z}_p$, the main difference between our approach and Schikhof's is that Schikhof requires the limit to exist for all $\mathfrak{z} \in \mathbb{Z}_p$ for which the quantity he denotes by $\mathcal{N}_\mu(\mathfrak{z})$ exists and is positive, whereas we do not. Indeed, in the case where $d\mu$ is the (p,q)-adic Haar measure, $\mathcal{N}_\mu(\mathfrak{z})$ exists and equals 1 for all $\mathfrak{z} \in \mathbb{Z}_p$ (this is given as an exercise in [19]'s appendix on integration), and so, applying Schikhof's notion of the derivative of a measure $d\mu$ with respect to the (p,q)-adic Haar measure would require the limit:

$$\lim_{N\to\infty} (D_N * d\mu)(\mathfrak{z}) \tag{2.146}$$

to exist in K *for all* $\mathfrak{z} \in \mathbb{Z}_p$. In contrast, *our* notion of Radon-Nikodym differentiation allows for the limit of the to fail to exist at some points.This distinction is significant.

In Part III of this series of papers, we will show that, for every odd prime number q, there exists a function $\hat{\chi}_q : \hat{\mathbb{Z}}_2 \to \overline{\mathbb{Q}}$ such that, after embedding $\overline{\mathbb{Q}}$ in $\mathbb{C}_q$, we have $\hat{\chi}_q \in B\left(\hat{\mathbb{Z}}_2, \mathbb{C}_q\right)$ and:

$$\lim_{N\to\infty} \sum_{|t|_2 \leq 2^N} \hat{\chi}_q(t)\, e^{2\pi i \{t\mathfrak{z}\}_2} \overset{\mathcal{F}_{2,q}}{=} \chi(\mathfrak{z}),\ \forall \mathfrak{z} \in \mathbb{Z}_2 \tag{2.147}$$

where the $\mathcal{F}_{2,q}$ means that the limit is taken in the topology of $\mathbb{Q}_q$ if $\mathfrak{z} \in \mathbb{Z}_2' = \mathbb{Z}_2 \backslash \mathbb{N}_0$ and is taken in the topology of $\mathbb{R}$ if $\mathfrak{z} \in \mathbb{N}_0$. Since $\hat{\chi}$ is q-adically bounded, the formula:

$$f \in C(\mathbb{Z}_2, \mathbb{C}_q) \mapsto \sum_{t \in \hat{\mathbb{Z}}_2} \hat{f}(t) \hat{\chi}_q(-t) \in \mathbb{C}_q \tag{2.148}$$

then defines a $(2,q)$-adic measure. Moreover, the convergence of the Fourier series (2.147) then formally justifies writing this measure as $\chi_q(\mathfrak{z})\,d\mathfrak{z}$. This is completely in line with the formal outline of Schikhof's Radon-Nikodym Theorem, save for the fact that the convergence of (2.147) does not occur q-adically for every $\mathfrak{z} \in \mathbb{Z}_2$.

In our (p,q)-adic context, Schikhof's result would be:

Theorem 14 (Schikhof's Radon-Nikodym Theorem [17]]). *Let $d\mu, d\nu \in C(\mathbb{Z}_p, K)'$, and suppose that the limit:*

$$g(\mathfrak{z}) \overset{def}{=} \lim_{N\to\infty} \frac{\int_{\mathfrak{z}+p^N\mathbb{Z}_p} d\nu(\mathfrak{x})}{\int_{\mathfrak{z}+p^N\mathbb{Z}_p} d\mu(\mathfrak{y})} \tag{2.149}$$

exists in K for all $\mathfrak{z} \in \mathbb{Z}_p$. Then, $g \in C(\mathbb{Z}_p, K)$ and:

$$\int_{\mathbb{Z}_p} f(\mathfrak{z})\,d\nu(\mathfrak{z}) = \int_{\mathbb{Z}_p} f(\mathfrak{z})\,g(\mathfrak{z})\,d\mu(\mathfrak{z}),\ \forall f \in C(\mathbb{Z}_p, K) \tag{2.150}$$

For χ_q, letting $d\nu$ be the measure whose Fourier-Stieltjes transform is $\hat{\chi}_q$, we have that:

$$\chi_q(\mathfrak{z}) \overset{\mathcal{F}_{2,q}}{=} \lim_{N\to\infty} \frac{\int_{\mathfrak{z}+2^N\mathbb{Z}_2} d\nu(\mathfrak{x})}{\int_{\mathfrak{z}+2^N\mathbb{Z}_2} d\mathfrak{y}},\ \forall \mathfrak{z} \in \mathbb{Z}_2 \tag{2.151}$$

where the limit is taken using the same convention as (2.147). Indeed, this limit of a ratio of integrals is precisely the same expression as (2.147). Where our approach diverges is that because $\chi_q(\mathfrak{z})$ is not $(2,q)$-adically continuous, we cannot make sense of the expression $\chi_q(\mathfrak{z})\,d\mathfrak{z}$ by interpreting it as the product of a continuous function against the (p,q)-adic Haar measure. Rather, we take the right-hand side from **Theorem 14**'s conclusion and use it to *define* what $g(\mathfrak{z})\,d\mu(\mathfrak{z})$ means in the context where $g(\mathfrak{z}) = \chi_q(\mathfrak{z})$ and $d\mu(\mathfrak{z}) = d\mathfrak{z}$.

2.6. **Exercises.** Throughout, we fix distinct prime numbers p and q, and let K be our non-archimedean q-adic field, except where explicitly stated otherwise.

Exercise 1. Let $\mathfrak{a} \in \mathbb{Z}_p$, let $f(\mathfrak{z}) \overset{\text{def}}{=} [\mathfrak{z} = \mathfrak{a}]$ be the indicator function of the point $\{\mathfrak{a}\}$ in $\mathbb{Z}_p$, viewed as taking values in $\mathbb{Q}_q$. Compute $S\{f\}(\mathfrak{z})$, the van der Put series of f evaluated at $\mathfrak{z}$. Discuss the convergence properties of $S\{f\}$. For what values of $\mathfrak{a}$ does $S\{f\} = f$?

Exercise 2. Letting q be an odd prime, recall from [24] that we can define $\chi_q : \mathbb{Z}_2 \to \mathbb{Z}_q$ as the unique solution of the functional equations:

$$\chi_q(2\mathfrak{z}) = \frac{\chi_q(\mathfrak{z})}{2} \tag{2.152}$$

$$\chi_q(2\mathfrak{z}+1) = \frac{q\chi_q(\mathfrak{z})+1}{2} \tag{2.153}$$

subject to the rising-continuity condition:

$$\lim_{n\to\infty} \chi_q([\mathfrak{z}]_{2^n}) \overset{\mathbb{Q}_q}{=} \chi(\mathfrak{z}),\ \forall \mathfrak{z} \in \mathbb{Z}_2 \tag{2.154}$$

Compute χ_q's van der Put series, $S\{\chi_q\}$. For what $\mathfrak{z} \in \mathbb{Z}_2$ is $S\{\chi_q\}(\mathfrak{z}) = \chi_q(\mathfrak{z})$? Describe the convergence of $S\{\chi_q\}(\mathfrak{z})$.

Exercise 3. Proceeding formally, compute:

$$\int_{\mathbb{Z}_2} \chi_q(\mathfrak{z})\, d\mathfrak{z} \tag{2.155}$$

and then use that result to compute:

$$\int_{\mathbb{Z}_2} \chi_q^2(\mathfrak{z})\, d\mathfrak{z} \tag{2.156}$$

Hint: Apply equation (2.97) from **Proposition 7**.

Exercise 4. Compute:

$$\sum_{t\in\hat{\mathbb{Z}}_p\backslash\{0\}} v_p(t)\, e^{2\pi i\{t\mathfrak{z}\}_p} \overset{\text{def}}{=} \lim_{N\to\infty} \sum_{0<|t|_p\leq p^N} v_p(t)\, e^{2\pi i\{t\mathfrak{z}\}_p} \tag{2.157}$$

For what $\mathfrak{z} \in \mathbb{Z}_p$ and which topological fields K does the limit exist? Describe the convergence of the limit at all $\mathfrak{z} \in \mathbb{Z}_p$ where the limit exists.

Exercise 5. Let $f \in C(\mathbb{Z}_p, K)$. How, if at all, could you define the integral:

$$\int_{\mathbb{Z}_p} f(\mathfrak{z})\, |\mathfrak{z}|_p^{-1}\, d\mathfrak{z}\ ? \tag{2.158}$$

Hint: Have you completed Exercise 4 yet?

Exercise 6. In this exercise, $d\mathfrak{z}$ denotes the real-valued Haar probability measure on $\mathbb{Z}_2$. Let q be an odd prime, and let $\hat{\varphi}_q : \hat{\mathbb{Z}}_q \to \mathbb{C}$ be defined by:

$$\hat{\varphi}_q(t) \overset{\text{def}}{=} \int_{\mathbb{Z}_2} e^{2\pi i\{t\chi_q(\mathfrak{z})\}_q} d\mathfrak{z}$$

Letting Σ be the Borel σ-algebra on $\mathbb{Z}_2$, let $\mathrm{P} : \Sigma \to [0,1]$ be the function defined by:

$$\mathrm{P}(E) \overset{\text{def}}{=} \int_{\mathbb{Z}_2} [\chi_q(\mathfrak{z}) \in E]\, d\mathfrak{z} \tag{2.159}$$

That is, $\mathrm{P}(E)$ is the Haar measure of $\chi_q^{-1}(E)$ in $\mathbb{Z}_2$. In particular, we write $\mathrm{P}\left(\chi_q \overset{q^n}{\equiv} k\right)$ to denote the real-valued Haar probability measure of the set of $\mathfrak{z} \in \mathbb{Z}_2$ at which $\chi_q(\mathfrak{z})$ is congruent to k mod q^n.

With all this, express $\mathrm{P}\left(\chi_q \overset{q^n}{\equiv} k\right)$ in terms of $\hat{\varphi}_q$. Then, viewing $[\chi_q]_{q^n}$ (the value attained by χ_q modulo q^n) as a random variable taking values in the set $\{0, \ldots, q^n - 1\}$, compute the expected value of $[\chi_q]_{q^n}$ in terms of $\hat{\varphi}_q$.

3. (p, q)-Adic Wiener Tauberian Theorems

Assumption 6. *THROUGHOUT THIS SECTION, UNLESS STATED OTHERWISE, WE ASSUME K IS NON-ARCHIMEDEAN.*

We begin with a useful notation for translations:

Definition 15. For $s \in \hat{\mathbb{Z}}_p$, $\mathfrak{a} \in \mathbb{Z}_p$, $\chi : \mathbb{Z}_p \to K$, and $\hat{\chi} : \hat{\mathbb{Z}}_p \to K$, we write:

$$\tau_s\{\hat{\chi}\}(t) \overset{\text{def}}{=} \hat{\chi}(t+s) \tag{3.1}$$

and:

$$\tau_{\mathfrak{a}}\{\chi\}(\mathfrak{z}) \overset{\text{def}}{=} \chi(\mathfrak{z}+\mathfrak{a}) \tag{3.2}$$

The proof of WTT for continuous (p,q)-adic functions is simple enough that it can be given directly without any preparatory work. For ease of reference, we restate the result before giving the proof.

Theorem 15 (Wiener Tauberian Theorem for Continuous (p,q)-adic Functions). *Let $\chi \in C(\mathbb{Z}_p, K)$. Then, the following are equivalent:*

I. $\frac{1}{\chi} \in C(\mathbb{Z}_p, K)$;

II. $\hat{\chi}$, *the Fourier transform, has a convolution inverse in* $c_0\left(\hat{\mathbb{Z}}_p, K\right)$.

III. The span of the translates of $\hat{\chi}$ *is dense in* $c_0\left(\hat{\mathbb{Z}}_p, K\right)$;

IV. χ *has no zeroes.*

Proof:

i. (I $\Rightarrow$ II) Suppose $1/\chi$ is continuous. Then, since the (p,q)-adic Fourier transform $\mathscr{F} : C(\mathbb{Z}_p, K) \to c_0\left(\hat{\mathbb{Z}}_p, K\right)$ is an isometric isomorphism of Banach algebras, it follows that:

$$\begin{gathered}\chi(\mathfrak{z}) \cdot \frac{1}{\chi(\mathfrak{z})} = 1,\ \forall \mathfrak{z} \in \mathbb{Z}_p \\ (\mathscr{F});\Updownarrow \\ \left(\hat{\chi} * \widehat{\left(\frac{1}{\chi}\right)}\right)(t) = \mathbf{1}_0(t),\ \forall t \in \hat{\mathbb{Z}}_p\end{gathered}$$

where both $\hat{\chi}$ and $\widehat{(1/\chi)}$ are in $c_0\left(\hat{\mathbb{Z}}_p, K\right)$. $\hat{\chi}$ has a convolution inverse in $c_0\left(\hat{\mathbb{Z}}_p, K\right)$, and this inverse is $\widehat{(1/\chi)}$.

ii. (II $\Rightarrow$ III) Suppose $\hat{\chi}$ has a convolution inverse $\hat{\chi}^{-1} \in c_0\left(\hat{\mathbb{Z}}_p, K\right)$. Then, letting $\hat{f} \in c_0\left(\hat{\mathbb{Z}}_p, K\right)$ be arbitrary, we have that:

$$\left(\hat{\chi} * \left(\hat{\chi}^{-1} * \hat{f}\right)\right)(t) = \left(\left(\hat{\chi} * \hat{\chi}^{-1}\right) * \hat{f}\right)(t) = \left(\mathbf{1}_0 * \hat{f}\right)(t) = \hat{f}(t),\ \forall t \in \hat{\mathbb{Z}}_p \tag{3.3}$$

In particular, letting $\hat{g}$ denote $\hat{\chi}^{-1} * \hat{f}$, we have that:

$$\hat{f}(t) = (\hat{\chi} * \hat{g})(t) \overset{K}{=} \lim_{N\to\infty} \sum_{|s|_p \le p^N} \hat{g}(s)\,\hat{\chi}(t-s) \tag{3.4}$$

Since $\hat{\chi}$ and $\hat{g}$ are in $c_0\left(\hat{\mathbb{Z}}_p, K\right)$, note that:

$$\begin{aligned}\sup_{t\in\hat{\mathbb{Z}}_p} \left| \sum_{s\in\hat{\mathbb{Z}}_p} \hat{g}(s)\,\hat{\chi}(t-s) - \sum_{|s|_p \le p^N} \hat{g}(s)\,\hat{\chi}(t-s) \right|_q &\le \sup_{t\in\hat{\mathbb{Z}}_p} \sup_{|s|_p > p^N} \left|\hat{g}(s)\,\hat{\chi}(t-s)\right|_q \\ \left(\|\hat{\chi}\|_{p^\infty,q} < \infty\right); &\le \sup_{|s|_p > p^N} \left|\hat{g}(s)\right|_q\end{aligned}$$

and hence:

$$\lim_{N\rightarrow\infty}\sup_{t\in\hat{\mathbb{Z}}_{p}}\left|\sum_{s\in\hat{\mathbb{Z}}_{p}}\hat{g}\left(s\right)\hat{\chi}\left(t-s\right)-\sum_{\left|s\right|_{p}\leq p^{N}}\hat{g}\left(s\right)\hat{\chi}\left(t-s\right)\right|_{q}\overset{\mathbb{R}}{=}\lim_{N\rightarrow\infty}\sup_{\left|s\right|_{p}>p^{N}}\left|\hat{g}\left(s\right)\right|_{q}\overset{\mathbb{R}}{=}0 \tag{3.5}$$

which shows that the q-adic convergence of $\sum_{\left|s\right|_{p}\leq p^{N}}\hat{g}\left(s\right)\hat{\chi}\left(t-s\right)$ to $\hat{f}\left(t\right)=\sum_{s\in\hat{\mathbb{Z}}_{p}}\hat{g}\left(s\right)\hat{\chi}\left(t-s\right)$ is uniform in t. Hence:

$$\lim_{N\rightarrow\infty}\sup_{t\in\hat{\mathbb{Z}}_{p}}\left|\hat{f}\left(t\right)-\sum_{\left|s\right|_{p}\leq p^{N}}\hat{g}\left(s\right)\hat{\chi}\left(t-s\right)\right|_{q}=0 \tag{3.6}$$

which is precisely what it means for the sequence $\left\{ \sum_{\left|s\right|_{p}\leq p^{N}}\hat{g}\left(s\right)\hat{\chi}\left(t-s\right)\right\} _{N\geq0}$ to converge in $c_{0}\left(\hat{\mathbb{Z}}_{p},K\right)$ (which has $\left\Vert \cdot\right\Vert _{p^{\infty},q}$ as its norm) to $\hat{f}$. Since $\hat{f}$ was arbitrary, and since this sequence is in the span of the translates of $\hat{\chi}$ over K, we see that said span is dense in $c_{0}\left(\hat{\mathbb{Z}}_{p},K\right)$.

iii. (III $\Rightarrow$ IV) Suppose $\text{span}_{K}\left\{ \tau_{s}\left\{ \hat{\chi}\right\} \left(t\right):s\in\hat{\mathbb{Z}}_{p}\right\}$ is dense in $c_{0}\left(\hat{\mathbb{Z}}_{p},K\right)$. Since $\mathbf{1}_{0}\left(t\right)\in c_{0}\left(\hat{\mathbb{Z}}_{p},K\right)$, given any $\epsilon\in\left(0,1\right)$, we can choose constants $\mathfrak{c}_{1},\ldots,\mathfrak{c}_{M}\in K$ and $t_{1},\ldots,t_{M}\in\hat{\mathbb{Z}}_{p}$ so that:

$$\sup_{t\in\hat{\mathbb{Z}}_{p}}\left|\mathbf{1}_{0}\left(t\right)-\sum_{m=1}^{M}\mathfrak{c}_{m}\hat{\chi}\left(t-t_{m}\right)\right|_{q}<\epsilon \tag{3.7}$$

Now, letting $N\geq\max\left\{ -v_{p}\left(t_{1}\right),\ldots,-v_{p}\left(t_{M}\right)\right\}$ be arbitrary (note that $-v_{p}\left(t\right)=-\infty$ when $t=0$), the maps $t\mapsto t+t_{m}$ are then bijections of the set $\left\{ t\in\hat{\mathbb{Z}}_{p}:\left|t\right|_{p}\leq p^{N}\right\}$. Consequently:

$$\begin{aligned}\sum_{\left|t\right|_{p}\leq p^{N}}\left(\sum_{m=1}^{M}\mathfrak{c}_{m}\hat{\chi}\left(t-t_{m}\right)\right)e^{2\pi i\left\{ t\mathfrak{z}\right\} _{p}} & =\sum_{m=1}^{M}\mathfrak{c}_{m}\sum_{\left|t\right|_{p}\leq p^{N}}\hat{\chi}\left(t\right)e^{2\pi i\left\{ \left(t+t_{m}\right)\mathfrak{z}\right\} _{p}}\\ & =\left(\sum_{m=1}^{M}\mathfrak{c}_{m}e^{2\pi i\left\{ t_{m}\mathfrak{z}\right\} _{p}}\right)\sum_{\left|t\right|_{p}\leq p^{N}}\hat{\chi}\left(t\right)e^{2\pi i\left\{ t\mathfrak{z}\right\} _{p}}\end{aligned}$$

Letting $N\rightarrow\infty$, we obtain:

$$\lim_{N\rightarrow\infty}\sum_{\left|t\right|_{p}\leq p^{N}}\left(\sum_{m=1}^{M}\mathfrak{c}_{m}\hat{\chi}\left(t-t_{m}\right)\right)e^{2\pi i\left\{ t\mathfrak{z}\right\} _{p}}\overset{K}{=}f_{m}\left(\mathfrak{z}\right)\chi\left(\mathfrak{z}\right) \tag{3.8}$$

where $f_{m}:\mathbb{Z}_{p}\rightarrow K$ is defined by:

$$f_{m}\left(\mathfrak{z}\right)\overset{\text{def}}{=}\sum_{m=1}^{M}\mathfrak{c}_{m}e^{2\pi i\left\{ t_{m}\mathfrak{z}\right\} _{p}},\ \forall\mathfrak{z}\in\mathbb{Z}_{p} \tag{3.9}$$

Moreover, this convergence is uniform with respect to $\mathfrak{z}$.

Consequently:

$$\begin{aligned}\left|1-f_{m}\left(\mathfrak{z}\right)\chi\left(\mathfrak{z}\right)\right|_{q} & \overset{\mathbb{R}}{=}\lim_{N\rightarrow\infty}\left|\sum_{\left|t\right|_{p}\leq p^{N}}\left(\mathbf{1}_{0}\left(t\right)-\sum_{m=1}^{M}\mathfrak{c}_{m}\hat{\chi}\left(t-t_{m}\right)\right)e^{2\pi i\left\{ t\mathfrak{z}\right\} _{p}}\right|_{q}\\
 & \leq\sup_{t\in\hat{\mathbb{Z}}_{p}}\left|\mathbf{1}_{0}\left(t\right)-\sum_{m=1}^{M}\mathfrak{c}_{m}\hat{\chi}\left(t-t_{m}\right)\right|_{q}\\
 & <\epsilon
\end{aligned}$$

for all $\mathfrak{z}\in\mathbb{Z}_{p}$.

If $\chi\left(\mathfrak{z}_{0}\right)=0$ for some $\mathfrak{z}_{0}\in\mathbb{Z}_{p}$, we would then have:

$$\epsilon>\left|1-f_{m}\left(\mathfrak{z}_{0}\right)\cdot0\right|_{q}=1 \tag{3.10}$$

which would contradict the fact that $\epsilon\in\left(0,1\right)$. As such, χ cannot have any zeroes whenever the span of $\hat{\chi}$'s translates are dense in $c_{0}\left(\hat{\mathbb{Z}}_{p},K\right)$.

iv. (IV $\Rightarrow$ I) Suppose χ has no zeroes. Since $\mathfrak{y}\mapsto1/\mathfrak{y}$ is a continuous map on $K\backslash\left\{ 0\right\}$, and since compositions of continuous maps are continuous, to show that $1/\chi$ is continuous, it suffices to show that $\left|\chi\left(\mathfrak{z}\right)\right|_{q}$ is bounded away from 0. To see this, suppose by way of contradiction that there was a sequence $\left\{ \mathfrak{z}_{n}\right\} _{n\geq0}\subseteq\mathbb{Z}_{p}$ such that for all $\epsilon>0$, $\left|\chi\left(\mathfrak{z}_{n}\right)\right|_{q}<\epsilon$ holds for all sufficiently large n—say, for all $n\geq N_{\epsilon}$. Since $\mathbb{Z}_{p}$ is compact, the $\mathfrak{z}_{n}$s have a subsequence $\mathfrak{z}_{n_{k}}$ which converges in $\mathbb{Z}_{p}$ to some limit $\mathfrak{z}_{\infty}\in\mathbb{Z}_{p}$. The continuity of χ then forces $\chi\left(\mathfrak{z}_{\infty}\right)=0$, but that contradicts the hypothesis that χ was given to have no zeroes.

Thus, if χ has no zeroes, $\left|\chi\left(\mathfrak{z}\right)\right|_{q}$ must be bounded away from zero, which then proves that $1/\chi$ is (p,q)-adically continuous.

Q.E.D.

The proof of WTT for (p,q)-adic measures is significantly more involved than the continuous case. While proving the non-vanishing of the limit of $\left(D_{N}*d\mu\right)\left(\mathfrak{z}\right)$ when $\hat{\mu}$'s translates have dense span is just as simple as in the continuous case, the other direction requires a much more intricate argument. The idea is this: by way of contradiction, suppose there is a $\mathfrak{z}_{0}\in\mathbb{Z}_{p}$ so that $\lim_{N\rightarrow\infty}\left(D_{N}*d\mu\right)\left(\mathfrak{z}_{0}\right)$ converges in K to zero, yet the span of $\hat{\mu}$'s translates is dense in $c_{0}\left(\hat{\mathbb{Z}}_{p},K\right)$. With these assumptions, we get a contradiction by constructing two functions $c_{0}\left(\hat{\mathbb{Z}}_{p},K\right)$ designed to produce certain behaviors when we convolve them with $\hat{\mu}$. The first function is constructed so as to give us something close to the constant function 0 (in the sense of $c_{0}\left(\hat{\mathbb{Z}}_{p},K\right)$'s norm, the (p,q)-adic sup norm on $\hat{\mathbb{Z}}_{p}$) when we convolve it with $\hat{\mu}$. On the other hand, our second function is constructed so as to give us something close to $\mathbf{1}_{0}$ when we convolve it with $\hat{\mu}$. By using the ultrametric inequality, we show these two estimates cannot *both* be true, yielding the desired contradiction.

While the idea is straightforward, the approach is somewhat technical because our argument will require restricting ourselves to working on particular neighborhoods of 0 in $\hat{\mathbb{Z}}_{p}$. The contradictory part of our estimates will emerge from delicate manipulation of convolutions in combination with these domain restrictions. Because of this, we need a notation for the supremum of the q-adic absolute value of function $\hat{\mathbb{Z}}_{p}\rightarrow K$ over a bounded neighborhood of 0.

Definition 16. For any integer $n \geq 1$ and any function $\hat{\chi} \in B\left(\hat{\mathbb{Z}}_p, K\right)$, we write $\|\hat{\chi}\|_{p^n,q}$ to denote:

$$\|\hat{\chi}\|_{p^n,q} \overset{\text{def}}{=} \sup_{|t|_p \leq p^n} |\hat{\chi}(t)|_q \tag{3.11}$$

This is a non-archimedean seminorm on $B\left(\hat{\mathbb{Z}}_p, K\right)$.

This convention explains why we write $\|\cdot\|_{p^\infty,q}$ to denote the supremum norm taken over all $t \in \hat{\mathbb{Z}}_p$: writing "$\|\cdot\|_{p,q}$" would potentially cause confusion with what we would write as $\|\cdot\|_{p^1,q}$.

While much of the intricate business involving the intermingling of restrictions and convolutions occurs in the various claims that structure our proof of **Theorem 3**, one particular result in this vein—the heart of **Theorem 3**'s proof—is non-trivial enough to merit separate consideration to avoid cluttering the flow of our argument. This is detailed in the following lemma.

Lemma 3. *Let $M, N \in \mathbb{N}_0$ be arbitrary, and let $\hat{\phi} \in B\left(\hat{\mathbb{Z}}_p, K\right)$ be supported on $|t|_p \leq p^N$. Then, for any $\hat{f}, \hat{g} : \hat{\mathbb{Z}}_p \to K$ where $\hat{g}$ is supported on $|t|_p \leq p^M$, we have:*

$$\left\|\hat{\phi} * \hat{f}\right\|_{p^M,q} \leq \left\|\hat{\phi}\right\|_{p^\infty,q} \left\|\hat{f}\right\|_{p^{\max\{M,N\}},q} \tag{3.12}$$

$$\left\|\hat{\phi} * \hat{f} * \hat{g}\right\|_{p^M,q} \leq \left\|\hat{\phi} * \hat{f}\right\|_{p^{\max\{M,N\}},q} \|\hat{g}\|_{p^\infty,q} \tag{3.13}$$

Proof: (3.12) is the easier of the two:

$$\begin{aligned}
\left\|\hat{\phi} * \hat{f}\right\|_{p^M,q} &= \sup_{|t|_p \leq p^M} \left| \sum_{s \in \hat{\mathbb{Z}}_p} \hat{\phi}(s) \hat{f}(t-s) \right|_q \\
\left(\hat{\phi}(s) = 0,\ \forall |s|_p > p^N\right); &\leq \sup_{|t|_p \leq p^M} \left| \sum_{|s|_p \leq p^N} \hat{\phi}(s) \hat{f}(t-s) \right|_q \\
\text{(ultrametric ineq.)}; &\leq \sup_{|t|_p \leq p^M} \sup_{|s|_p \leq p^N} \left| \hat{\phi}(s) \hat{f}(t-s) \right|_q \\
&\leq \sup_{|s|_p \leq p^N} \left|\hat{\phi}(s)\right|_q \sup_{|t|_p \leq p^{\max\{M,N\}}} \left|\hat{f}(t)\right|_q \\
&= \left\|\hat{\phi}\right\|_{p^\infty,q} \cdot \left\|\hat{f}\right\|_{p^{\max\{M,N\}},q}
\end{aligned}$$

and we are done.

Proving (3.13) is similar, but more involved. We start by writing out the convolution of $\hat{\phi} * \hat{f}$ and $\hat{g}$:

$$\begin{aligned}
\left\|\hat{\phi} * \hat{f} * \hat{g}\right\|_{p^M,q} &= \sup_{|t|_p \le p^M} \left|\sum_{s \in \hat{\mathbb{Z}}_p} \left(\hat{\phi} * \hat{f}\right)(t-s)\,\hat{g}(s)\right|_q \\
&\le \sup_{|t|_p \le p^M} \sup_{s \in \hat{\mathbb{Z}}_p} \left|\left(\hat{\phi} * \hat{f}\right)(t-s)\,\hat{g}(s)\right|_q \\
\left(\hat{g}(s) = 0,\ \forall |s|_p > p^M\right);\ &\le \sup_{|t|_p \le p^M} \sup_{|s|_p \le p^M} \left|\left(\hat{\phi} * \hat{f}\right)(t-s)\,\hat{g}(s)\right|_q \\
&\le \|\hat{g}\|_{p^\infty,q} \sup_{|t|_p,|s|_p \le p^M} \left|\left(\hat{\phi} * \hat{f}\right)(t-s)\right|_q \\
\left(\text{write out } \hat{\phi} * \hat{f}\right);\ &= \|\hat{g}\|_{p^\infty,q} \sup_{|t|_p,|s|_p \le p^M} \left|\sum_{\tau \in \hat{\mathbb{Z}}_p} \hat{\phi}(t-s-\tau)\,\hat{f}(\tau)\right|_q \\
(\text{let } u = s+\tau);\ &= \|\hat{g}\|_{p^\infty,q} \sup_{|t|_p,|s|_p \le p^M} \left|\sum_{u-s \in \hat{\mathbb{Z}}_p} \hat{\phi}(t-u)\,\hat{f}(u-s)\right|_q \\
\left(s + \hat{\mathbb{Z}}_p = \hat{\mathbb{Z}}_p\right);\ &= \|\hat{g}\|_{p^\infty,q} \sup_{|t|_p,|s|_p \le p^M} \left|\sum_{u \in \hat{\mathbb{Z}}_p} \hat{\phi}(t-u)\,\hat{f}(u-s)\right|_q
\end{aligned}$$

Now we use the fact that $\hat{\phi}(t-u)$ vanishes for all $|t-u|_p > p^N$. Because t is restricted to $|t|_p \le p^M$, observe that for $|u|_p > p^{\max\{M,N\}}$, the ultrametric inequality allows us to write:

$$|t-u|_p = \max\left\{|t|_p, |u|_p\right\} > p^{\max\{M,N\}} > p^N \tag{3.14}$$

So, for $|t|_p, |s|_p \le p^M$, the summand $\hat{\phi}(t-u)\,\hat{f}(u-s)$ vanishes whenever $|u|_p > p^{\max\{M,N\}}$. This gives us:

$$\left\|\hat{\phi} * \hat{f} * \hat{g}\right\|_{p^M,q} \le \|\hat{g}\|_{p^\infty,q} \sup_{|t|_p,|s|_p \le p^M} \left|\sum_{|u|_p \le p^{\max\{M,N\}}} \hat{\phi}(t-u)\,\hat{f}(u-s)\right|_q \tag{3.15}$$

Next, we expand the range of $|s|_p$ and $|t|_p$ from $\le p^M$ to $\le p^{\max\{M,N\}}$:

$$\left\|\hat{\phi} * \hat{f} * \hat{g}\right\|_{p^M,q} \le \|\hat{g}\|_{p^\infty,q} \sup_{|t|_p,|s|_p \le p^{\max\{M,N\}}} \left|\sum_{|u|_p \le p^{\max\{M,N\}}} \hat{\phi}(t-u)\,\hat{f}(u-s)\right|_q \tag{3.16}$$

In doing so, note that we have put s, t, and u into a single p-adic neighborhood of 0:

$$\left\{x \in \hat{\mathbb{Z}}_p : |x|_p \le p^{\max\{M,N\}}\right\} \tag{3.17}$$

This is good, because this neighborhood is closed under addition; for any $\left|s\right|_{p}\leq p^{\max\{M,N\}}$, our u-sum is invariant under the change of variables $u\mapsto u+s$, and we obtain:

$$\begin{aligned}\left\Vert \hat{\phi}*\hat{f}*\hat{g}\right\Vert _{p^{M},q} & \leq\left\Vert \hat{g}\right\Vert _{p^{\infty},q}\sup_{\left|t\right|_{p},\left|s\right|_{p}\leq p^{\max\{M,N\}}}\left|\sum_{\left|u\right|_{p}\leq p^{\max\{M,N\}}}\hat{\phi}\left(t-\left(u+s\right)\right)\hat{f}\left(u\right)\right|_{q}\\
 & =\left\Vert \hat{g}\right\Vert _{p^{\infty},q}\sup_{\left|t\right|_{p},\left|s\right|_{p}\leq p^{\max\{M,N\}}}\left|\sum_{\left|u\right|_{p}\leq p^{\max\{M,N\}}}\hat{\phi}\left(t-s-u\right)\hat{f}\left(u\right)\right|_{q}
\end{aligned}$$

Finally, observing that:

$$\left\{ t-s:\left|t\right|_{p},\left|s\right|_{p}\leq p^{\max\{M,N\}}\right\} =\left\{ t:\left|t\right|_{p}\leq p^{\max\{M,N\}}\right\} \tag{3.18}$$

we can write:

$$\begin{aligned}\left\Vert \hat{\phi}*\hat{f}*\hat{g}\right\Vert _{p^{M},q} & \leq\left\Vert \hat{g}\right\Vert _{p^{\infty},q}\sup_{\left|t\right|_{p}\leq p^{\max\{M,N\}}}\left|\sum_{\left|u\right|_{p}\leq p^{\max\{M,N\}}}\hat{\phi}\left(t-u\right)\hat{f}\left(u\right)\right|_{q}\\
 & \leq\left\Vert \hat{g}\right\Vert _{p^{\infty},q}\sup_{\left|t\right|_{p}\leq p^{\max\{M,N\}}}\left|\underbrace{\sum_{u\in\hat{\mathbb{Z}}_{p}}\hat{\phi}\left(t-u\right)\hat{f}\left(u\right)}_{\hat{\phi}*\hat{f}}\right|_{q}\\
 & =\left\Vert \hat{g}\right\Vert _{p^{\infty},q}\sup_{\left|t\right|_{p}\leq p^{\max\{M,N\}}}\left|\left(\hat{\phi}*\hat{f}\right)\left(u\right)\right|_{q}\\
\left(\text{by definition}\right); & =\left\Vert \hat{g}\right\Vert _{p^{\infty},q}\left\Vert \hat{\phi}*\hat{f}\right\Vert _{p^{\max\{M,N\}},q}
\end{aligned}$$

This proves the desired estimate, and with it, the rest of the **Lemma**.

Q.E.D.

Now, we can prove **Theorem 3**.

Theorem 16 (**Wiener Tauberian Theorem for (p,q)-adic Measures**). *Let $d\mu\in C\left(\mathbb{Z}_{p},K\right)^{\prime}$. Then, the span of the translates of the Fourier-Stieltjes transform of $d\mu$ is dense in $c_{0}\left(\hat{\mathbb{Z}}_{p},K\right)$ if and only if:*

$$\lim_{N\rightarrow\infty}\left(D_{N}*d\mu\right)\left(\mathfrak{z}\right)$$

is non-zero for all $\mathfrak{z}\in\mathbb{Z}_{p}$ at which the limit exists in K.

Proof:

Note: For brevity, we write $\tilde{\mu}_{N}$ to denote $D_{N}*d\mu$.

We start with the simpler of the two directions.

I. Suppose the span of the translates of $\hat{\mu}$ are dense. Just as in the proof of the WTT for continuous (p,q)-adic functions, we let $\epsilon \in (0,1)$ and then choose $\mathfrak{c}_m$s and t_ms so that:

$$\sup_{t\in\hat{\mathbb{Z}}_p}\left|\mathbf{1}_0(t)-\sum_{m=1}^{M}\mathfrak{c}_m\hat{\mu}(t-t_m)\right|_q<\epsilon \tag{3.19}$$

Picking sufficiently large N, we obtain:

$$\begin{aligned}\left|1-\left(\sum_{m=1}^{M}\mathfrak{c}_m e^{2\pi i\{t_m\mathfrak{z}\}_p}\right)\tilde{\mu}_N(\mathfrak{z})\right|_q &\leq \max_{|t|_p\leq p^N}\left|\mathbf{1}_0(t)-\sum_{m=1}^{M}\mathfrak{c}_m\hat{\mu}(t-t_m)\right|_q\\ &\leq \sup_{t\in\hat{\mathbb{Z}}_p}\left|\mathbf{1}_0(t)-\sum_{m=1}^{M}\mathfrak{c}_m\hat{\mu}(t-t_m)\right|_q\\ &<\epsilon\end{aligned}$$

Now, let $\mathfrak{z}_0\in\mathbb{Z}_p$ be a point so that $\mathfrak{L}\overset{\text{def}}{=}\lim_{N\rightarrow\infty}\tilde{\mu}_N(\mathfrak{z}_0)$ converges in K. We need to show $\mathfrak{L}\neq 0$. To do this, plugging $\mathfrak{z}=\mathfrak{z}_0$ into the above yields:

$$\epsilon>\lim_{N\rightarrow\infty}\left|1-\left(\sum_{m=1}^{M}\mathfrak{c}_m e^{2\pi i\{t_m\mathfrak{z}_0\}_p}\right)\tilde{\mu}_N(\mathfrak{z}_0)\right|_q=\left|1-\left(\sum_{m=1}^{M}\mathfrak{c}_m e^{2\pi i\{t_m\mathfrak{z}_0\}_p}\right)\cdot\mathfrak{L}\right|_q \tag{3.20}$$

If $\mathfrak{L}=0$, the right-most expression will be 1, and hence, we get $\epsilon>1$, but this is impossible; ϵ was given to be less than 1. So, if $\lim_{N\rightarrow\infty}\tilde{\mu}_N(\mathfrak{z}_0)$ converges in K to $\mathfrak{L}$, $\mathfrak{L}$ must be non-zero.

II. Let $\mathfrak{z}_0\in\mathbb{Z}_p$ be a zero of $\tilde{\mu}(\mathfrak{z})$ such that $\tilde{\mu}_N(\mathfrak{z}_0)\rightarrow 0$ in K as $N\rightarrow\infty$. Then, by way of contradiction, suppose the span of the translates of $\hat{\mu}$ is dense in $c_0\left(\hat{\mathbb{Z}}_p,K\right)$, despite the zero of $\tilde{\mu}$ at $\mathfrak{z}_0$. Thus, we can use linear combinations of translates of $\hat{\mu}$ approximate any function in $c_0\left(\hat{\mathbb{Z}}_p,K\right)$'s sup norm. In particular, we choose to approximate $\mathbf{1}_0$: letting $\epsilon\in(0,1)$ be arbitrary, there is then a choice of $\mathfrak{c}_j$s in K and t_js in $\hat{\mathbb{Z}}_p$ so that:

$$\sup_{t\in\hat{\mathbb{Z}}_p}\left|\mathbf{1}_0(t)-\sum_{j=1}^{J}\mathfrak{c}_j\hat{\mu}(t-t_j)\right|_q<\epsilon \tag{3.21}$$

Letting:

$$\hat{\eta}_\epsilon(t)\overset{\text{def}}{=}\sum_{j=1}^{J}\mathfrak{c}_j\mathbf{1}_{t_j}(t) \tag{3.22}$$

we can express the above linear combination as the convolution:

$$(\hat{\mu}*\hat{\eta}_\epsilon)(t)=\sum_{\tau\in\hat{\mathbb{Z}}_p}\hat{\mu}(t-\tau)\sum_{j=1}^{J}\mathfrak{c}_j\mathbf{1}_{t_j}(\tau)=\sum_{j=1}^{J}\mathfrak{c}_j\hat{\mu}(t-t_j) \tag{3.23}$$

So, we get:

$$\|\mathbf{1}_0-\hat{\mu}*\hat{\eta}_\epsilon\|_{p^\infty,q}\overset{\text{def}}{=}\sup_{t\in\hat{\mathbb{Z}}_p}|\mathbf{1}_0(t)-(\hat{\mu}*\hat{\eta}_\epsilon)(t)|_q<\epsilon \tag{3.24}$$

Before proceeding any further, it is vital to note that we can (and must) assume that $\hat{\eta}_\epsilon$ is not identically 0; this must hold whenever $\epsilon\in(0,1)$, because—were $\hat{\eta}_\epsilon$ identically zero—the supremum:

$$\sup_{t\in\hat{\mathbb{Z}}_p}\left|\mathbf{1}_0(t)-(\hat{\mu}*\hat{\eta}_\epsilon)(t)\right|_q$$

would then be equal to 1, rather than $<\epsilon$.

That being done, equation (3.24) shows the assumption we want to contradict: the existence of a function which produces something close to $\mathbf{1}_0$ after convolution with $\hat{\mu}$. We will arrive at our contradiction by showing that the zero of $\tilde{\mu}$ at $\mathfrak{z}_0$ allows us to construct a second function which yields something close to 0 after convolution with $\hat{\mu}$. By convolving *both* of these functions with $\hat{\mu}$, we'll end up with something which is close to both $\mathbf{1}_0$ *and* close to 0, which is, of course, impossible.

Our make-things-close-to-zero-by-convolution function is going to be:

$$\hat{\phi}_N(t)\overset{\text{def}}{=}\mathbf{1}_0\left(p^Nt\right)e^{-2\pi i\{t\mathfrak{z}_0\}_p},\ \forall t\in\hat{\mathbb{Z}}_p \tag{3.25}$$

Note that $\hat{\phi}_N(t)$ is only supported for $|t|_p\leq p^N$. Now, as defined, we have that:

$$\begin{aligned}\left(\hat{\mu}*\hat{\phi}_N\right)(\tau)&=\sum_{s\in\hat{\mathbb{Z}}_p}\hat{\mu}(\tau-s)\hat{\phi}_N(s)\\ \left(\hat{\phi}_N(s)=0,\ \forall|s|_p>p^N\right);&=\sum_{|s|_p\leq p^N}\hat{\mu}(\tau-s)\hat{\phi}_N(s)\\ &=\sum_{|s|_p\leq p^N}\hat{\mu}(\tau-s)e^{-2\pi i\{s\mathfrak{z}_0\}_p}\end{aligned}$$

Fixing τ, observe that the map $s\mapsto\tau-s$ is a bijection of the set $\left\{s\in\hat{\mathbb{Z}}_p:|s|_p\leq p^N\right\}$ whenever $|\tau|_p\leq p^N$. So, for any $N\geq -v_p(\tau)$, we obtain:

$$\begin{aligned}\left(\hat{\mu}*\hat{\phi}_N\right)(\tau)&=\sum_{|s|_p\leq p^N}\hat{\mu}(\tau-s)e^{-2\pi i\{s\mathfrak{z}_0\}_p}\\ &=\sum_{|s|_p\leq p^N}\hat{\mu}(s)e^{-2\pi i\{(\tau-s)\mathfrak{z}_0\}_p}\\ &=e^{-2\pi i\{\tau\mathfrak{z}_0\}_p}\sum_{|s|_p\leq p^N}\hat{\mu}(s)e^{2\pi i\{s\mathfrak{z}_0\}_p}\\ &=e^{-2\pi i\{\tau\mathfrak{z}_0\}_p}\tilde{\mu}_N(\mathfrak{z}_0)\end{aligned}$$

Since this holds for all $N\geq -v_p(\tau)$, upon letting $N\to\infty$, $\tilde{\mu}_N(\mathfrak{z}_0)$ converges to 0 in K, by our assumption. Thus, for any $\epsilon'>0$, there exists an $N_{\epsilon'}$ so that $|\tilde{\mu}_N(\mathfrak{z}_0)|_q<\epsilon'$ for all $N\geq N_{\epsilon'}$. Combining this with the above computation (after taking q-adic absolute values), we have established the following:

Claim 2. Let $\epsilon'>0$ be arbitrary. Then, there exists an $N_{\epsilon'}\geq 0$ (depending only on $\hat{\mu}$ and ϵ') so that, for all $\tau\in\hat{\mathbb{Z}}_p$:

$$\left|\left(\hat{\mu}*\hat{\phi}_N\right)(\tau)\right|_q=\left|e^{-2\pi i\{\tau\mathfrak{z}_0\}_p}\tilde{\mu}_N(\mathfrak{z}_0)\right|_q<\epsilon',\ \forall N\geq\max\left\{N_{\epsilon'},-v_p(\tau)\right\},\tau\in\hat{\mathbb{Z}}_p \tag{3.26}$$

As stated, the idea is to convolve $\hat{\mu}*\hat{\phi}_N$ with $\hat{\eta}_\epsilon$ so as to obtain a function (via the associativity of convolution) which is both close to 0 and close to $\mathbf{1}_0$. However, our present situation is less than ideal because the lower bound on N in **Claim 2** depends on τ, and so, the convergence of $\left|\left(\hat{\mu}*\hat{\phi}_N\right)(\tau)\right|_q$ to 0 as $N\rightarrow\infty$ will *not* be uniform in τ. This is where the difficulty of this direction of the proof lies. To overcome this obstacle, instead of convolving $\hat{\mu}*\hat{\phi}_N$ with $\hat{\eta}_\epsilon$, we will convolve $\hat{\mu}*\hat{\phi}_N$ with a truncated version of $\hat{\eta}_\epsilon$, whose support has been restricted to a finite subset of $\hat{\mathbb{Z}}_p$. This is the function $\hat{\eta}_{\epsilon,M}:\hat{\mathbb{Z}}_p\rightarrow K$ given by:

$$\hat{\eta}_{\epsilon,M}(t)\overset{\textrm{def}}{=}\mathbf{1}_0\left(p^M t\right)\hat{\eta}_\epsilon(t)=\begin{cases}\hat{\eta}_\epsilon(t) & \textrm{if } \left|t\right|_p\leq p^M\\ 0 & \textrm{else}\end{cases}\tag{3.27}$$

where $M\geq 0$ is arbitrary. With this, we can attain the "close to both 0 and $\mathbf{1}_0$" contradiction we desire for $\hat{\mu}*\hat{\phi}_N*\hat{\eta}_{\epsilon,M}$. The proof will be complete once we've demonstrated that this contradiction will persist even as $M\rightarrow\infty$.

The analogue of the estimate (3.24) for this truncated case is:

$$\begin{aligned}\left(\hat{\mu}*\hat{\eta}_{\epsilon,M}\right)(t)&=\sum_{\tau\in\hat{\mathbb{Z}}_p}\hat{\mu}(t-\tau)\mathbf{1}_0\left(p^M\tau\right)\sum_{j=1}^{J}\mathfrak{c}_j\mathbf{1}_{t_j}(\tau)\\&=\sum_{\left|\tau\right|_p\leq p^M}\hat{\mu}(t-\tau)\sum_{j=1}^{J}\mathfrak{c}_j\mathbf{1}_{t_j}(\tau)\\&=\sum_{j:\left|t_j\right|_p\leq p^M}\mathfrak{c}_j\hat{\mu}(t-t_j)\end{aligned}$$

where, *note*, instead of summing over all the t_js that came with $\hat{\eta}_\epsilon$, we only sum over those t_js in the set $\left\{t\in\hat{\mathbb{Z}}_p:\left|t\right|_p\leq p^M\right\}$. Because the t_js came with the un-truncated $\hat{\eta}_\epsilon$, observe that we can make $\hat{\mu}*\hat{\eta}_{\epsilon,M}$ *equal* to $\hat{\mu}*\hat{\eta}_\epsilon$ by simply choosing M to be large enough so that all the t_js lie in $\left\{t\in\hat{\mathbb{Z}}_p:\left|t\right|_p\leq p^M\right\}$. The lower bound for this choice of M is:

$$M_0\overset{\textrm{def}}{=}\max\left\{-v_p(t_1),\ldots,-v_p(t_J)\right\}\tag{3.28}$$

Then, we have:

$$\left(\hat{\mu}*\hat{\eta}_{\epsilon,M}\right)(t)=\sum_{j=1}^{J}\mathfrak{c}_j\hat{\mu}(t-t_j)=\left(\hat{\mu}*\hat{\eta}_\epsilon\right)(t),\ \forall M\geq M_0,\ \forall t\in\hat{\mathbb{Z}}_p\tag{3.29}$$

So, applying the $\left\|\cdot\right\|_{p^M,q}$ semi-norm:

$$\begin{aligned}\left\|\mathbf{1}_0-\hat{\mu}*\hat{\eta}_{\epsilon,M}\right\|_{p^M,q}&=\sup_{\left|t\right|_p\leq p^M}\left|\mathbf{1}_0(t)-\left(\hat{\mu}*\hat{\eta}_{\epsilon,M}\right)(t)\right|_q\\(\textrm{if } M\geq M_0);&=\sup_{\left|t\right|_p\leq p^M}\left|\mathbf{1}_0(t)-\left(\hat{\mu}*\hat{\eta}_\epsilon\right)(t)\right|_q\\&\leq\left\|\mathbf{1}_0-\left(\hat{\mu}*\hat{\eta}_\epsilon\right)\right\|_{p^\infty,q}\\&<\epsilon\end{aligned}$$

Finally, note that M_0 depends on *only* $\hat{\eta}_\epsilon$ and ϵ. **Claim 3**, given below, summarizes these findings:

Claim 3. Let $\epsilon>0$ be arbitrary. Then, there exists an integer M_0 depending only on $\hat{\eta}_\epsilon$ and ϵ, so that:

I. $\hat{\mu}*\hat{\eta}_{\epsilon,M}=\hat{\mu}*\hat{\eta}_\epsilon,\ \forall M\geq M_0$.

II. $\left\|\mathbf{1}_0-\hat{\mu}*\hat{\eta}_{\epsilon,M}\right\|_{p^{\infty},q}<\epsilon,\ \forall M\geq M_0$.

III. $\left\|\mathbf{1}_0-\hat{\mu}*\hat{\eta}_{\epsilon,M}\right\|_{p^{M},q}<\epsilon,\ \forall M\geq M_0$.

The next step is to refine our "make $\hat{\mu}$ close to zero" estimate by taking into account $\left\|\cdot\right\|_{p^m,q}$.

Claim 4. Let $\epsilon'>0$. Then, there exists $N_{\epsilon'}$ depending only on ϵ' and $\hat{\mu}$ so that:

$$\left\|\hat{\phi}_N*\hat{\mu}\right\|_{p^m,q}<\epsilon',\ \forall m\geq 1,\ \forall N\geq\max\left\{N_{\epsilon'},m\right\}\tag{3.30}$$

Proof of claim: Let $\epsilon'>0$. **Claim 2** tells us that there is an $N_{\epsilon'}$ (depending on ϵ', $\hat{\mu}$) so that:

$$\left|\left(\hat{\phi}_N*\hat{\mu}\right)(\tau)\right|_q<\epsilon',\ \forall N\geq\max\left\{N_{\epsilon'},-v_p(\tau)\right\},\ \forall\tau\in\hat{\mathbb{Z}}_p\tag{3.31}$$

So, letting $m\geq 1$ be arbitrary, note that $\left|\tau\right|_p\leq p^m$ implies $-v_p(\tau)\leq m$. As such, we can make the result of **Claim 2** hold for all $\left|\tau\right|_p\leq p^m$ by choosing $N\geq\max\left\{N_{\epsilon'},m\right\}$:

$$\underbrace{\sup_{\left|\tau\right|_p\leq p^m}\left|\left(\hat{\phi}_N*\hat{\mu}\right)(\tau)\right|_q}_{\left\|\hat{\mu}*\hat{\phi}_N\right\|_{p^m,q}}<\epsilon',\ \forall N\geq\max\left\{N_{\epsilon'},m\right\}\tag{3.32}$$

This proves the claim.

Using **Lemma 3**, we can now set up the string of estimates we need to arrive at the desired contradiction. First, let us choose an $\epsilon\in(0,1)$ and a function $\hat{\eta}_\epsilon:\hat{\mathbb{Z}}_p\to K$ which is not identically zero, so that:

$$\left\|\mathbf{1}_0-\hat{\mu}*\hat{\eta}_\epsilon\right\|_{p^{\infty},q}<\epsilon\tag{3.33}$$

Then, by **Claim 3**, we can choose a M_ϵ depending only on ϵ and $\hat{\eta}_\epsilon$ so that:

$$M\geq M_\epsilon\Rightarrow\left\|\mathbf{1}_0-\hat{\mu}*\hat{\eta}_{\epsilon,M}\right\|_{p^M,q}<\epsilon\tag{3.34}$$

This shows that $\hat{\mu}$ can be convolved to become close to $\mathbf{1}_0$. The contradiction will follow from convolving this with $\hat{\phi}_N$, where N, at this point, is arbitrary:

$$\left\|\hat{\phi}_N*\left(\mathbf{1}_0-\left(\hat{\mu}*\hat{\eta}_{\epsilon,M}\right)\right)\right\|_{p^M,q}=\left\|\hat{\phi}_N-\left(\hat{\phi}_N*\hat{\mu}*\hat{\eta}_{\epsilon,M}\right)\right\|_{p^M,q}\tag{3.35}$$

Our goal here is to show that (3.35) is close to both 0 and 1 simultaneously.

First, writing:

$$\left\|\hat{\phi}_N*\left(\mathbf{1}_0-\left(\hat{\mu}*\hat{\eta}_{\epsilon,M}\right)\right)\right\|_{p^M,q}\tag{3.36}$$

the fact that $\hat{\phi}_N(t)$ is supported on $\left|t\right|_p\leq p^N$ allows us to apply equation (3.12) from **Lemma 3**. This gives the estimate:

$$\left\|\hat{\phi}_N*\left(\mathbf{1}_0-\left(\hat{\mu}*\hat{\eta}_{\epsilon,M}\right)\right)\right\|_{p^M,q}\leq\underbrace{\left\|\hat{\phi}_N\right\|_{p^{\infty},q}}_{1}\left\|\mathbf{1}_0-\left(\hat{\mu}*\hat{\eta}_{\epsilon,M}\right)\right\|_{p^{\max\{M,N\}},q}\tag{3.37}$$

for all M and N. Letting $M \geq M_\epsilon$, we can apply **Claim 3** and write $\hat{\mu} * \hat{\eta}_{\epsilon,M} = \hat{\mu} * \hat{\eta}_\epsilon$. So, for $M \geq M_\epsilon$ and arbitrary N, we have:

$$\begin{aligned} \left\| \hat{\phi}_N * (\mathbf{1}_0 - (\hat{\mu} * \hat{\eta}_{\epsilon,M})) \right\|_{p^M,q} &\leq \left\| \mathbf{1}_0 - (\hat{\mu} * \hat{\eta}_\epsilon) \right\|_{p^{\max\{M,N\}},q} \\ &\leq \left\| \mathbf{1}_0 - (\hat{\mu} * \hat{\eta}_\epsilon) \right\|_{p^\infty,q} \\ (\textbf{Claim } 3)\,; &< \epsilon \end{aligned}$$

Thus, the *left-hand side* of (3.35) is $< \epsilon$. This is the first estimate.

Keeping $M \geq M_\epsilon$ and N arbitrary, we will obtain the desired contradiction by showing that the *right-hand side* of (3.35) is $> \epsilon$. Since $\|\cdot\|_{p^M,q}$ is a non-archimedean semi-norm, it satisfies the ultrametric inequality. Applying this to the right-hand side of (3.35) yields:

$$\left\| \hat{\phi}_N - \left(\hat{\phi}_N * \hat{\mu} * \hat{\eta}_{\epsilon,M} \right) \right\|_{p^M,q} \leq \max \left\{ \left\| \hat{\phi}_N \right\|_{p^M,q}, \left\| \hat{\phi}_N * \hat{\mu} * \hat{\eta}_{\epsilon,M} \right\|_{p^M,q} \right\} \tag{3.38}$$

Because $\hat{\eta}_{\epsilon,M}(t)$ and $\hat{\phi}_N$ are supported on $|t|_p \leq p^M$ and $|t|_p \leq p^N$, respectively, we can apply (3.13) from **Lemma 3** and write:

$$\left\| \hat{\phi}_N * \hat{\mu} * \hat{\eta}_{\epsilon,M} \right\|_{p^M,q} \leq \left\| \hat{\phi}_N * \hat{\mu} \right\|_{p^{\max\{M,N\}},q} \cdot \left\| \hat{\eta}_\epsilon \right\|_{p^\infty,q} \tag{3.39}$$

Since $\hat{\eta}_\epsilon$ was given to *not* be identically zero, the quantity $\|\hat{\eta}_\epsilon\|_{p^\infty,q}$ must be positive. Consequently, for our given $\epsilon \in (0,1)$, let us define ϵ' by:

$$\epsilon' = \frac{\epsilon}{2 \left\| \hat{\eta}_\epsilon \right\|_{p^\infty,q}} \tag{3.40}$$

So far, N is still arbitrary. By **Claim 4**, for this ϵ', we know there exists an $N_{\epsilon'}$ (depending only on ϵ, $\hat{\eta}_\epsilon$, and $\hat{\mu}$) so that:

$$\left\| \hat{\phi}_N * \hat{\mu} \right\|_{p^m,q} < \epsilon', \ \forall m \geq 1, \ \forall N \geq \max\{N_{\epsilon'}, m\} \tag{3.41}$$

Choosing $m = N_{\epsilon'}$ gives us:

$$N \geq N_{\epsilon'} \Rightarrow \left\| \hat{\phi}_N * \hat{\mu} \right\|_{p^N,q} < \epsilon' \tag{3.42}$$

So, choose $N \geq \max\{N_{\epsilon'}, M\}$. Then, $\max\{M,N\} = N$, and so (3.39) becomes:

$$\left\| \hat{\phi}_N * \hat{\mu} * \hat{\eta}_{\epsilon,M} \right\|_{p^M,q} \leq \left\| \hat{\phi}_N * \hat{\mu} \right\|_{p^N,q} \cdot \left\| \hat{\eta}_\epsilon \right\|_{p^\infty,q} < \frac{\epsilon}{2 \left\| \hat{\eta}_\epsilon \right\|_{p^\infty,q}} \cdot \left\| \hat{\eta}_\epsilon \right\|_{p^\infty,q} = \frac{\epsilon}{2} \tag{3.43}$$

Since $\left\| \hat{\phi}_N \right\|_{p^M,q} = 1$ for all $M, N \geq 0$, this shows that:

$$\left\| \hat{\phi}_N * \hat{\mu} * \hat{\eta}_{\epsilon,M} \right\|_{p^M,q} < \frac{\epsilon}{2} < 1 = \left\| \hat{\phi}_N \right\|_{p^M,q} \tag{3.44}$$

Now comes the hammer: by the ultrametric inequality, (3.38) is an *equality* whenever one of $\left\| \hat{\phi}_N \right\|_{p^M,q}$ or $\left\| \hat{\phi}_N * \hat{\mu} * \hat{\eta}_{\epsilon,M} \right\|_{p^M,q}$ is strictly greater than the other. Having proved that to be the case, (3.35) becomes:

$$\epsilon > \left\| \hat{\phi}_N * (\mathbf{1}_0 - (\hat{\mu} * \hat{\eta}_{\epsilon,M})) \right\|_{p^M,q} = \left\| \hat{\phi}_N - \left(\hat{\phi}_N * \hat{\mu} * \hat{\eta}_{\epsilon,M} \right) \right\|_{p^M,q} = 1 > \epsilon \tag{3.45}$$

for all $M \geq M_\epsilon$ and all $N \geq \max\{N'_\epsilon, M\}$. The left-hand side is our first estimate, and the right-hand side is our second. Since $\epsilon < 1$, this is clearly impossible.

This proves that the existence of the zero $\mathfrak{z}_0$ precludes the translates of $\hat{\mu}$ from being dense in $c_0\left(\hat{\mathbb{Z}}_p, K\right)$.

Q.E.D.

As a note on future work, the author intends to pursue (p,q)-adic Fourier Analysis and Wiener Tauberian Theorems in the context of functions on $\mathbb{Q}_p$ and metrically complete algebraic extensions thereof in future work, though not in papers in this series. Unpublished independent research by the author indicates that, at the formal level, the resulting theory is virtually identical to the one Vladimirov presented in his paper on real- and complex-valued distributions and generalized functions of a p-adic variable [27].

4. Solutions to Exercises

Solution 1. If $\mathfrak{a} \notin \mathbb{N}_0$, $S\{f\}$ is identically 0, and is therefore not equal to f. If $\mathfrak{a} \in \mathbb{N}_0$, $S\{f\}$ is given by:

$$S\{f\}(\mathfrak{z}) = \left[\mathfrak{z} \overset{p^{\lambda_p(\mathfrak{a})}}{\equiv} \mathfrak{a}\right] - \sum_{n=0}^{\infty}\sum_{j=1}^{p-1}\left[\mathfrak{z} \overset{p^{\lambda_p(\mathfrak{a})+n+1}}{\equiv} \mathfrak{a} + jp^{\lambda_p(\mathfrak{a})+n}\right] \tag{4.1}$$

and $S\{f\} = f$ everywhere. In this case, the series converges point-wise with respect to $\mathfrak{z}$.

Solution 2. The van der Put series of χ_q is:

$$\sum_{n=1}^{\infty} \frac{q^{\#_1(n)-1}}{2^{\lambda_2(n)}}\left[\mathfrak{z} \overset{2^{\lambda_2(n)}}{\equiv} n\right] \tag{4.2}$$

This series represents χ_q everywhere and converges q-adically to χ_q point-wise with respect to $\mathfrak{z}$.

Solution 3. We have:

$$\int_{\mathbb{Z}_2} \chi_q(\mathfrak{z})\, d\mathfrak{z} = -\frac{1}{q-3} \tag{4.3}$$

$$\int_{\mathbb{Z}_2} \chi_q^2(\mathfrak{z})\, d\mathfrak{z} = \frac{q+3}{(q-3)(q^2-7)} \tag{4.4}$$

Solution 4. For all $\mathfrak{z} \in \mathbb{Z}_p \backslash \{0\}$:

$$\sum_{t \in \hat{\mathbb{Z}}_p \backslash \{0\}} v_p(t)\, e^{2\pi i \{t\mathfrak{z}\}_p} = \frac{p\,|\mathfrak{z}|_p^{-1} - 1}{p-1} \tag{4.5}$$

The series converges point-wise with respect to $\mathfrak{z}$. In particular:

$$\sum_{0<|t|_p \leq p^N} v_p(t)\, e^{2\pi i \{t\mathfrak{z}\}_p} = \frac{p\,|\mathfrak{z}|_p^{-1} - 1}{p-1} \tag{4.6}$$

whenever $N > v_p(\mathfrak{z})$. Since the series converges point-wise in finitely many steps, the series converges in the discrete topology, and thus makes sense in any topological field of characteristic $\neq p$.

Solution 5. Rearranging the result from **Exercise 4**, we have:

$$|\mathfrak{z}|_p^{-1} = \frac{1}{p} + \sum_{t \in \hat{\mathbb{Z}}_p \backslash \{0\}} \frac{p-1}{p} v_p(t)\, e^{2\pi i \{t\mathfrak{z}\}_p} \tag{4.7}$$

Since $p \neq q$, the function $\hat{\mu} : \hat{\mathbb{Z}}_p \to K$ defined by:

$$\hat{\mu}(t) \overset{\text{def}}{=} \begin{cases} \frac{1}{p} & \text{if } t = 0 \\ \frac{p-1}{p} v_p(t) & \text{else} \end{cases} \tag{4.8}$$

is then an element of $B\left(\hat{\mathbb{Z}}_p, K\right)$ which is, in a sense, the Fourier-Stieltjes transform of $\left|\mathfrak{z}\right|_p^{-1}$. Thus, for any $f \in C(\mathbb{Z}_p, K)$, we can interpret the integral:

$$\int_{\mathbb{Z}_p} f(\mathfrak{z}) \left|\mathfrak{z}\right|_p^{-1} d\mathfrak{z} \tag{4.9}$$

as the image of f under this measure, and so, by the Parseval-Plancherel formula:

$$\begin{aligned} \int_{\mathbb{Z}_p} f(\mathfrak{z}) \left|\mathfrak{z}\right|_p^{-1} d\mathfrak{z} &= \sum_{t \in \hat{\mathbb{Z}}_p} \hat{f}(t) \hat{\mu}(-t) \\ &= \frac{\hat{f}(0)}{p} + \frac{p-1}{p} \sum_{t \in \hat{\mathbb{Z}}_p \setminus \{0\}} \hat{f}(t) v_p(-t) \hat{\mu}(-t) \end{aligned}$$

It is worth noting that if $K = \mathbb{C}$, and $f : \mathbb{Z}_p \to \mathbb{C}$ is represented by an absolutely convergent Fourier series, this formula will be equal to the integral $\int_{\mathbb{Z}_p} f(\mathfrak{z}) \left|\mathfrak{z}\right|_p^{-1} d\mathfrak{z}$ in the sense of [27].

Solution 6. We have:

$$\mathrm{P}\left(\chi_q \overset{q^n}{\equiv} k\right) = \frac{1}{q^n} \sum_{\left|t\right|_q \leq q^n} \hat{\varphi}_q(t) e^{-2\pi i t k} \tag{4.10}$$

Hence:

$$\mathbb{E}\left(\left[\chi_q\right]_{q^n}\right) = \sum_{k=0}^{q^n-1} k \mathrm{P}\left(\chi_q \overset{q^n}{\equiv} k\right) = \frac{q^n - 1}{2} - \sum_{0 < \left|t\right|_q \leq q^n} \frac{\hat{\varphi}_q(t)}{1 - e^{-2\pi i t}} \tag{4.11}$$

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Declaration of Interests

- The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

- The author declares the following financial interests/personal relationships which may be considered as potential competing interests: None.

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